LMLinga: A WEB-BASED HOUSEHOLD PROFILING AND AUTOMATED SPOT-MAPPING SYSTEM WITH A MULTILINGUAL CHATBOT

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CHAPTER 1
INTRODUCTION

This chapter discusses the background of the study and the current difficulties in overseeing medical records and services at the Barangay Health Center of La Medalla. It provides a brief overview of the study's goals, significance, scope, and important terms while presenting LMLinga as a suggested digital solution to these problems.

1.1 Project Context

Barangay Health Centers (BHCs) play an important role in maintaining residents' health and well-being. They are the ones who provide health services to the residents, especially to those who lack the financial means to receive treatment at a hospital [9]. To improve the delivery of these services, many BHCs utilize technology such as health information systems, automated generation of reports, and mobile/web applications [3, 5, 11, 15]. These technologies help improve the performance of health centers in terms of paperwork processing, accuracy of health data, and efficiency of health monitoring. This demonstrates that technology plays a significant part in enhancing the overall operation of Barangay Health Centers.

The World Health Organization (WHO) actively promotes the use of Information Technology in community health centers to enhance universal health coverage through electronic medical records (EMR), telemedicine, and the Internet of Things (IoT), among other technologies [21]. In line with this, international studies indicate that IT adoption contributes to improvements in efficiency, effectiveness, productivity, and quality of health services while also enhancing communication, collaboration, and recognition among health workers [12, 20]. Information systems allow health workers to record and manage data accurately, and the generation of reports becomes faster and can be accessed in real-time [12], but this will only be effective if the health workers have the capabilities to use the technology [18]. Due to this, the use of user-friendly interfaces can greatly help the health workers to use IT efficiently and confidently.

Similarly, the Department of Health (DOH) pushes Universal Health Care, and the adoption of information technology in barangay health centers has surged through the use of mobile applications like eHATID, now used in over 450 LGUs [22], Community Health Tracking System (CHITS) [23], and BHW Connect [24]. These IT adoptions remarkably supported and improved barangay health services in the areas of networking, infrastructure, office automation, and development and implementation of computer-based systems. These developments demonstrate that information technology has significantly improved barangay health center services, paving the way for more efficient and effective community health care.

The La Medalla Barangay Health Center in Iriga, Camarines Sur, Bicol, Philippines, is a small health-care unit catering to a population of 2,878 individuals living in 651 households. The health center is staffed by 10 health workers and is divided into three groups. Barangay Health Workers (BHWs) conduct house-to-house profiling and assist in basic health monitoring. Barangay Nutrition Scholars (BNSs) focus on child immunization, nutrition monitoring, deworming, and vitamin supplementation. Barangay Service Point Officers (BSPOs) are responsible for the implementation of family planning services. All health workers are supervised by the head midwife, who provides consultations and medical guidance when the resident is experiencing health-related problems.

Currently, most of their processes still rely on manual and paper-based records. During house-to-house profiling, Barangay Health Workers (BHWs) carry heavy and bulky papers to record the information of each household, which must later be sorted manually by zone for report generation that will be submitted to the City Health Office. The health center also uses a spot map that displays the barangay and the locations of households. The map includes predefined health indicators, helping the health workers to locate the households more easily. However, the household locations are mainly plotted based on the memory of the health workers, so household locations are sometimes unreliable. These manual processes limit efficiency and accuracy, highlighting the need for a more streamlined and reliable system to support barangay health services.

The reliance on these manual and paper-based processes at Barangay La Medalla Health Center highlights several problems and challenges that hinder the efficient delivery of health services. BHWs experience time-consuming, physically tiring tasks that are highly prone to human errors in data collection and organization. They also experience delays in report generation due to limited manpower. Storing records is prone to damage because they are kept on paper. As health centers depend on paper-based record-keeping, information retrieval from hard copies is a challenge; they are subject to delays in service delivery and data duplication.

In addition to these challenges, the health center's spot map is prone to human error because households are plotted from memory, making the accuracy of the households' locations unreliable. On the other hand, residents also experience difficulties in accessing important health information due to busy schedules, physical distance, and mobility issues. As a result, residents have limited access to timely, reliable, and accurate health information, which may affect their awareness of proper health practices and preventive measures. These issues highlight the need for a more effective, precise, and user-friendly system to handle data, optimize processes, and guarantee residents' and healthcare professionals' fast and dependable access to health information.

To solve these existing problems, the researchers propose LMLinga, a web-based household profiling and automated spot mapping system with a chatbot. LMLinga will transfer the current manual processes in the Barangay Health Center of La Medalla into a streamlined digital format, enabling BHWs to collect, organize, manage, and view household records without the need for bulky logbooks during house-to-house visits. The automated spot mapping process will automatically plot the household on a digital map with predefined health indicators, ensuring data accuracy for enhanced health monitoring. The chatbot will enhance community awareness by serving as a digital front desk, providing residents with reliable health information. The implementation of LMLinga will enable health workers to save time and reduce workloads, increasing the time available for direct patient care and community health services.

1.2 Purpose and Description of the Project

The purpose of the study is to design and develop LMLinga, a web-based household profiling and automated spot-mapping system with a multilingual chatbot. LMLinga will allow the Barangay Health Workers (BHWs) to use a digital tool to streamline community health data entry monitoring and improve resident access to preventive health information. The study focuses on the Barangay Health Center of La Medalla to promote efficient workflows, data accuracy for health workers to prioritize care over paperwork, and health information accessibility for the residents.

The system will have five main features that aim to improve the health services of Barangay La Medalla Health Center. The system will provide digital household profiling, which replaces paper forms with a centralized database where health workers can easily record and manage household information. It includes automated report generation that quickly generates collected data into summary reports, reducing the time needed for manual reporting. The system also has an interactive spot map using Leaflet.js, which displays the collected household coordinates on a digital map to help health workers monitor household locations in the barangay. It also features a multilingual chatbot powered by RAG and Ollama that can provide health information to residents in English, Tagalog, and the local Bikol-Iriga dialect. To protect the resident data and ensure privacy, the system will use role-based access control (RBAC).

1.3 Objectives of the Project

The general objective of this study is to design and develop LMLinga, a web-based household profiling and automated spot-mapping system with a multilingual chatbot. The system aims to digitize the manual and paper-based record management at the Barangay Health Center of La Medalla to enhance workflow efficiency, data accuracy, and health information accessibility for both residents and health workers. Specifically, this project aims to achieve the following:

1. To collect and evaluate user stories and technical acceptance criteria under the Extreme Programming framework to establish the functional requirements for transitioning the manual household profiling, spot-mapping, and record-keeping operations at the Barangay Health Center in La Medalla into a centralized system.
2. To design and develop LMLinga, a web-based system that uses geographic data to automate spot mapping, digitize household profiling, and integrate a multilingual chatbot that provides residents with basic health information.
3. To conduct systematic testing and assessment to verify that each completed module satisfies user requirements using:

   3.1. User Acceptance Testing

   3.2. Formative Usability Testing

   3.3. Integration Testing

   3.4. Regression Testing

   3.5. Security Testing
4. To deploy LMLinga in the Barangay Health Center of La Medalla and assess how well it improves data accuracy, workflow efficiency, and health information accessibility for both residents and health workers.

1.4 Significance of the Study

This study aims to develop LMLinga, a web-based household profiling system and automated spot mapping with a chatbot for the Barangay Health Center of La Medalla. Through the implementation of LMLinga, the study seeks to improve the management of health data, accessibility of information, and efficiency of health services of the barangay. With this, the study is significant to the following:

Barangay Health Center. This research has great importance to the La Medalla Barangay Health Center as it helps in managing health records in an organized and secure manner through digital means. The health center will be able to avoid problems such as duplication of data, damage to data, and delays in accessing information through the elimination of paper-based documentation.

Residents of the Barangay. This research will benefit the residents as they will have better access to health information through the chatbot. They will no longer have to go to the health center to get information about health programs and IEC materials from DOH health brochures. This will be particularly beneficial for senior residents and those who have mobility issues. The residents will also have better awareness about preventive health practices.

Camarines Sur Polytechnic Colleges. The study is important to Camarines Sur Polytechnic Colleges as it shows the application of information technology in solving real-world problems. The study will highlight how students can create innovative solutions using information technology that can be aligned with the goals of sustainable development.

Researchers. The study has great relevance for researchers because it helped them to apply and gain more knowledge and skills in creating a web-based household profile, automated spot mapping, and multilingual chatbots for LMLinga to resolve actual problems faced by the Barangay Health Center of La Medalla.

Future Researchers. The proposed study can be used as a reference for future researchers who are planning to develop similar web-based health management systems. The proposed study provides a foundation for further improvements in digital profiling, GIS-based monitoring, and chatbot integration in primary healthcare settings.

1.5 Scope and Limitations of the Project

The purpose of this study is to develop LMLinga, a web-based household profiling and automated spot-mapping with a multilingual chatbot, which will be used by the Barangay Health Center of La Medalla and its authorized barangay officials to address the current issues in manual household profiling, paper-based records, and hand-drawn maps. The system will include features such as digital household profiling, automated spot-mapping for visualizing household locations, secure data management, automated report generation, and a multilingual chatbot to assist residents, and the project will follow the Agile Software Development Methodology for iterative development and improvement. Also, the system will be developed using modern web technologies and mapping tools like Leaflet.js, RAG, and Ollama, and is expected to be completed within the project implementation of the study. To make sure that the system will work effectively, it will be evaluated based on its functionality, usability, and security through testing and feedback from the Barangay Health Workers. Throughout this process, the study aims to provide a reliable digital solution that improves the efficiency of the health data management and monitoring of Barangay La Medalla.

The system is limited to the Barangay Health Center of La Medalla and is not intended for use in other barangays or integration with official government health databases. While the system includes a multilingual chatbot, it is limited to providing general, verified health information and cannot diagnose diseases, prescribe treatment, or replace professional medical consultation, as its purpose is limited only to information dissemination. The chatbot does not have access to individual household health records and only provides general health information. Access to resident health information is restricted to verified household head representatives and authorized personnel based on the system-assigned roles to ensure data privacy and security. The multilingual chatbot supports only English, Tagalog, and Bikol-Iriga languages. Although the system allows offline data capture and synchronization, full functionality, such as real-time monitoring, chatbot responses, plotting of households' locations on the digital map, and report generation, requires a stable internet connection. These limitations define the system’s boundaries in terms of coverage, functionality, accessibility, and technical capability within the intended development scope.

1.6 Project Dictionary

The section presents the key terms and concepts used in this study to ensure clarity and consistent understanding throughout the manuscript. It includes both conceptual definitions, which are based on credible sources, and operational definitions, which explain how each term is specifically applied within the LMLinga system. This section helps readers clearly understand the technical and research-related terms used in the project.

Agile Software Development. It refers to an iterative and flexible software development method that facilitates collaboration among team members, coupled with the ability to continually improve throughout the entire software development process, allows development teams to produce functional software in a timely manner while responding appropriately to an ever-changing set of user requirements [16]. In this study, Agile refers to the development methodology used in creating LMLinga, where the system is developed in phases with continuous feedback from Barangay Health Workers of La Medalla to improve features such as profiling, mapping and report generation.

Artificial Intelligence (AI). It refers to the ability of computers to accomplish precisely the same things that humans do each day, such as learning, reasoning, problem-solving or decision-making, as well as generating responses from available data. It involves the development of intelligent technologies to analyze data, recognize patterns and ultimately support automated actions or decisions [7]. In this study, AI refers to the chatbot feature of LMLinga that automatically responds to user inquiries using programmed logic and predefined health information.

Automated. It refers to a technology-based solution that operates according to a set of predetermined rules, automatically executing tasks with little or no human involvement. It enables activities to be performed automatically, increasing efficiency, reliability, and precision, and reducing the need for manual labor [17]. In this study, automated refers to system processes that are performed automatically by LMLinga without manual computation, such as generating reports and plotting household data once information is entered into the system.

Chatbot. It refers to computer software that can replicate human conversation by using written or oral dialogue and describes a computer software application designed to interactively emulate human communication with a system. The program allowed the user to naturally and interactively communicate with the program by interpreting user input and generating a response, thus providing the user with information, assistance, and support [19]. In this study, the chatbot serves as a multilingual digital front desk for the Barangay Health Workers of La Medalla, providing health-related information.

Geographic Information System (GIS). It refers to a set of computer-based tools used to collect, save, maintain, examine, and graphically represent data that is referenced to geographic locations. GIS is a combination of both spatial and non-spatial data that aids in the creation of maps, performing spatial analyses, and making decisions by allowing users to better understand patterns, relationships, and trends in relation to geographic locations [14]. In this study, GIS refers to the mapping technology integrated into LMLinga to digitally plot and monitor household locations within Barangay La Medalla.

Health Center. It refers to a community-based primary healthcare facility that provides comprehensive health services, including preventive, promotive, diagnostic, and curative care. It is designed to be a source of access to patient-centered healthcare services for all; particularly for the underserved populations, and can provide a full array of medical, dental, mental and social services to the community [10]. In this study, health center refers to the Barangay Health Center of Barangay La Medalla, which serves as the source of health-related information, consultations, and records that are encoded, managed, and accessed through the LMLinga system for profiling and reporting purposes.

Household. It refers to an individual or a person living alone or a group of individuals a family living together, who share common living arrangements or resources in the same housing unit. It represents the primary unit of analysis for the collection and organization of demographic, socioeconomic, and social or community-related data [25]. In this study, a household refers to a family or group of individuals living in the same residence within Barangay La Medalla who's demographic and health-related information are recorded and managed in the LMLinga system.

Profiling. It refers to the systematic collection, organization, and analysis of group data or personal data, with the aim of finding the characteristics, conditions, and patterns of data that can be used to assist in the classification, assessment, and decision-making. The process creates structured profiles from collected information for monitoring, planning, and providing service delivery [4]. In this study, profiling refers to the process of collecting, organizing, and maintaining household and resident information in the LMLinga system to support health monitoring, reporting, and services.

Local Storage. It refers to data stored within the user's browser for web applications using key-value pairs and remains persistent beyond a single session. The stored data will remain available until the end user completes the process of deleting or removing it, or the application removes it [26]. In this study, local storage refers to the feature that temporarily saves collected household data when there is no internet connection and automatically syncs once connectivity is restored.

Multilingual. It refers to the capability of any individual, tool, or system to utilize, comprehend, and communicate with two or more distinct languages. It involves the ability to work with and process a variety of languages, so that it is possible to communicate effectively between users of different languages [13]. In this study, multilingual refers to the capability of the LMLingua chatbot to communicate and provide health-related information in both English and Tagalog to accommodate different users in Barangay La Medalla.

Spot-Mapping. It refers to the use of a geographic information system (GIS) to depict the location of specific items or occurrences as spots on a map for the purpose of evaluating the geographic distribution or pattern of those items, as well as their interaction with each other. It is commonly used to provide a method of visually representing geographic data and is widely used in support of spatial analysis and decision-making [6]. In this study, Spot-Mapping refers to the feature of LMLinga that digitally displays households as clickable markers on a map of Barangay La Medalla, allowing Barangay Health Workers to monitor and view resident health information based on location.

RAG. It refers to a method for producing text from an artificial intelligence approach that combines large language models with a system of information retrieval. By using both large language models and external data sources, this approach generates more accurate responses, produces high-quality, context-aware responses, and produces factually correct responses by retrieving and integrating relevant and current information [1]. In this study, Retrieval-Augmented Generation (RAG) refers to the process used in the LMLinga system where user queries from the multilingual chatbot are processed by retrieving relevant information from a structured database and integrating it into generated responses.

System. It refers to a collection of interconnected components or elements working in harmony to fulfil a predetermined function or objective. It is characterized by relationships between its parts, where each component works together to produce the combined results of the entire system [8]. In this study, the system refers to the LMLinga web-based platform that integrates household profiling, automated spot mapping, report generation, and a multilingual chatbot to help Barangay Health Workers manage and monitor community health data efficiently.

Web-based Application. It refers to a software program that runs on a web server and is available through a browser via the internet. It describes all types of technology that use the web infrastructure to function and deliver; therefore, users can use these systems from remote locations without having to install the software on their device locally [2]. In this study, the web-based application refers to LMLinga, which can be accessed through smartphones and desktop computers for profiling, mapping, reporting, and chatbot interaction.

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CHAPTER 2
REVIEW OF RELATED LITERATURE AND SYSTEMS

The chapter provides an extensive examination of both local and international research studies and existing systems, which establish the core foundation for developing LMLinga. The section draws on established theories, technological trends, and prior empirical research to establish a problem context that identifies essential methods and tools.

2.1 Related Literature and Systems

The section presents relevant concepts, published research, and existing software applications to establish the technical and conceptual foundations of the proposed system. The review is organized topically, covering the topics needed to support the proposed system.

2.1.1 Barangay Health Centers and Digital Transformation in Primary Healthcare

The Philippines' first point of contact with its community for basic medical/ preventive health services is through a barangay health center (BHC). BHCs offer a variety of important services, including immunization, maternal and child health care, nutrition, family planning, and routine monitoring of health with the assistance of barangay health workers (BHW) and midwives who supervise the staff, etc. [37]. As communities' demand for health care keeps increasing, many health care facilities are starting to utilize digital health technologies to improve health care delivery, monitoring of patients, and data management [16]. By implementing integrated healthcare information systems, local health facilities are more effectively organizing patients' records, generating reports faster, and increasing access to patient information for both the health care team and the patient.

Various new studies show that the implementation of digital healthcare systems has improved the efficiency of operations and enhanced the delivery of services in primary care settings. For example, Elepaño et al. [2025] provide evidence that the introduction of electronic health records has improved the organization, retrieval, and management of health records in primary care facilities in the Philippines and has facilitated the coordination of care among health workers [16]. In addition, Gonzaga Ballaran et al. [2023] provide further evidence of an improved ability for Local Government Units to manage their residents through barangay profiling systems that provide a centralized digital database and automate reporting processes, thus reducing the amount of time spent entering data into a manual system and increasing the veracity and accessibility of residents' records [19]. As such, the adoption of digital platforms provides health workers with more time to deliver healthcare services rather than having to dedicate time to processing paper records.

Furthermore, healthcare information systems are increasingly being created to facilitate local healthcare delivery by ensuring these systems can be made accessible, usable, and adaptable. For instance, Sucaldito et al. [2025] have indicated that the adoption of healthcare technologies in the Philippines' primary care sector is enhanced when the system design aligns with the healthcare workers' operational workflow and technological capabilities [35]. Features like simplified user interfaces, offline access capabilities, multilingual support, and centralized data management enhance the overall effectiveness of community healthcare. The aforementioned technologies improve the ability of healthcare workers to manage healthcare records efficiently, as well as enable an uninterrupted provision of health services to the community.

The reviewed studies illustrate that the operation of modern healthcare information systems has advanced through the use of digital records, centralized databases, automated report generation, and interactive monitoring technologies. Consequently, the findings from these studies are relevant to the current research study as they provide both a technological and operational foundation from which to establish the LMLinga system as a Centralized Healthcare Information System for Barangay La Medalla. By providing digital household profiles, automated spot mapping, multilingual chatbot assistance, and centralized record management, the proposed system aims to that these hurdles can be overcome through user-centric design and the integration of automated improve the organization of healthcare data, provide greater access to healthcare information, and facilitate the efficient delivery of healthcare services to the community.

2.1.2 Extreme Programming (XP) Requirements Elicitation and Specification Modeling

In Agile software engineering, requirements collection and specification modeling emphasize flexibility, ongoing partnership, and incremental refinement instead of creating comprehensive requirements documentation that is fixed before starting development. Agile methods support software development via the division of software projects into smaller development iterations, allowing for continued improvements in system functionality as feedback is received from users [15]. This practice works particularly well with healthcare information systems since business processes and end-user requests may change at any time throughout a project's lifespan. Agile methods, such as Extreme Programming (XP), allow for the use of low-overhead user-centric means of collecting user requirements (e.g., user stories and acceptance criteria) rather than depending on large volumes of requirements documentation that were developed at the start of a project. These types of approaches help developers to see the day-to-day tasks and operational requirements of end users (providers) prior to having to implement major system capabilities.

Collaboration among stakeholders is an important principle of development using XP methodologies. In a healthcare institution, as users are involved throughout the development process, the end user will feel increased usability as well as acceptance of the system. As explained by Elepaño et al. [2025] in their study of health information systems, active participation by healthcare workers in both system development and evaluation would lead to increased effectiveness of such systems within healthcare institutions [16]. In addition to these studies, the rationale behind involving end users in their development is that frequent feedback from these end users will allow developers to detect problems such as operational issues, interface problems, and workflow discrepancies early in the development cycle. A similar finding was reported by Sucaldito et al. [2025], who indicated that local primary care health workers in the Philippines experience numerous issues with usability, digital literacy skills, and Internet connectivity issues when attempting to adopt digital health systems [35]. Of critical importance is the advantage of iteratively developing and revising systems for use in community health operations. By using XP methods in developing new software features, BHWs will have the ability to modify those features to meet practical demands and technical expertise.

The specification modeling mentioned above for XP helps to quantify user needs regarding system functionality through establishing functional and non-functional requirements. Thus, functional requirements describe how well an XP product performs; that is to say, what operational tasks the XP system has to accomplish, e.g., household profiling, automated report generation, GIS-based spot mapping, and chatbot-assisted information retrieval [19]. For example, in the context of the LMLinga system, functional requirements include the storing of resident information, validating health records, generating demographic reports, and displaying where residents live via interactive digital maps. Non-functional requirements define the quality and/or operational constraints imposed upon the system, e.g., security, usability, reliability, maintainability, and performance efficiency [17]. In healthcare activities, protecting the privacy of resident health information is critical while making it possible for healthcare workers to access and use resident data in resource-constrained circumstances.

Agile and XP development methodologies also support continuous testing and validation throughout the software development lifecycle. Systematic evaluation and continuous quality assurance (QA) are critical components of healthcare Information Systems due to the sensitive and mission-critical nature of the data that is handled by these systems [17]. Through frequent iterative test cycles and ongoing stakeholder feedback, developers can quickly identify problems with database consistency, interface usability, or workflow integration before these issues cause negative effects on the entire system architecture. The iterative validation process of continuous testing facilitates the mitigation of development risks such as scope creep, inconsistent data structures, and mismatch of features. The incremental delivery structure of XP allows the developer to enhance individual modules of the system while still maintaining the stability and maintainability of the entire system.

XP methodologies are especially applicable to developing localized healthcare systems because community health settings typically require adaptable and flexible software solutions. Continuous refinement of features such as automated spot mapping, multilingual chatbot responses, and household profiling will take place through the iterative elicitation of requirements and specification modelling while soliciting direct input from Barangay Health Workers and other health professionals within the province of La Medalla. Utilization of this development methodology will ensure that the LMLinga system remains in alignment with the real-world health functional activities and operational conditions that occur in Barangay La Medalla while sustaining usability, dependability, and accuracy of data.

Agile and Extreme Programming (XP) methods enhance software development's ability to improve by continually collaborating with stakeholders, iteratively gathering requirements, and frequently refining systems in response to evolving business requirements. The literature indicates that involving healthcare providers when developing a system is critical in ensuring that healthcare information systems can perform reliably, maintain usability, and conform to the actual operational workflows of the healthcare professionals utilizing them. Furthermore, the literature indicates that defining both functional and non-functional requirements through continuous feedback and testing can mitigate the risks associated with software development, enhance system quality, and ultimately lead to more effective data management for the respective healthcare environments. Consequently, the findings in this literature form the foundational basis for developing LMLinga as a centralized healthcare information system for Barangay La Medalla. By utilizing XP principles, the proposed study will effectively gather user stories, define technical acceptance criteria, and questions. This study emphasizes that general-purpose AI like ChatGPT requires local medical professional review and country-specific training for optimal effectiveness in adhering to regional regulations and protocols. Next, the Filipino ASR Bot focuses on assessing children's physical health and iteratively refining system features (i.e., household profile creation, automated spot mapping, multilingual chatbots, and secure record management) according to the needs of Barangay Health Workers and healthcare services.

2.1.3 Household Profiling and Web-Based Health Information Systems

Household profiling is a systematic approach that involves gathering, organizing, and managing the demographic and health-related information of families within a community to facilitate efficient planning, governance, and service distribution [19]. This process allows local administrators and health workers to understand household composition and community needs, making it easier to identify priority groups and plan appropriate services. Traditionally, barangay health workers relied on recording household data manually using paper-based forms, which often resulted in delays, data redundancy, and difficulties in tracking residents over time [20]. These manual methods make it challenging to maintain accurate and updated records, particularly when information must be retrieved frequently for reporting or monitoring purposes. By transitioning to a web-based profiling system, these procedures become more accurate and efficient since the digital platform can securely store resident information, automate report generation, and provide analytical tools that support decision-making [20]. As a result, household information can be accessed and updated more efficiently, improving the organization and reliability of community data.

These innovations allow barangay officials and health workers to retrieve household information instantly, monitor health indicators at the family level, and identify priority areas for interventions [35]. This capability strengthens the ability of local health personnel to observe trends and respond more quickly to community health concerns. However, determining challenges such as fragmented data, management, limited interoperability, and varying levels of digital literacy affect the effectiveness of health information systems in primary care settings [35]. These barriers highlight the importance of designing systems that are user-friendly and adaptable to the capabilities of health workers. Recent evaluations of Philippine barangay health offices suggest that these hurdles can be overcome through user-centric design and the integration of automated reporting, which reduces the physical and cognitive load on health workers [14]. Thus, according to Rae et al. [2025], a well-designed web-based system can address these limitations by enhancing information governance, improving access to reliable data, and improving evidence-based planning and policy development [31]. Together, these studies suggest that integrating household profiling with a web-based health information system can enhance community-level health monitoring, streamline reporting processes, and promote data-driven decision-making in barangay health centers.

Literature related to household profiling and web-based health information systems supports the idea that centralized digital platforms will serve as a replacement for paper records, which in turn will assist in improving organizing, retrieving, and reporting data. Other literature demonstrates that centralized digital platforms increase data accuracy, improve workflow efficiency, and enhance access to Health Information. These findings support the development and implementation of the LMLinga system at the Barangay Health Center in La Medalla.

2.1.4 Geographic Information Systems (GIS) and Automated Spot Mapping in Healthcare

Geographic Information Systems (GIS) represent revolutionary digital platforms in healthcare that allow for the spatial analysis of demographic trends, illness patterns, and environmental dangers [18]. This technology provides the essential components to transfer the current mapping system of the Barangay Health Center of La Medalla, which relies on health workers' memory of traditional hand-drawn maps, to an automated spot-mapping system. Unlike the traditional hand-drawn maps, GIS can handle vast amounts of geospatial data and present it in a visually accessible format, which is essential for decision-making [12]. By integrating this technology into a web-based system, the process of spot mapping becomes automated, plotting household locations and health indicators in a digital map for more accurate health monitoring. Unlike static paper maps that require manual redrawing to reflect community changes, an Interactive mapping is also an important feature for improving the manual spot-mapping process currently used in many local health centers. According to Marie [2026], Leaflet.js is a lightweight automated GIS allows for dynamic updates. As soon as a Barangay Health Worker (BHW) inputs new household data, the digital spot map reflects these changes instantly.

Automated spot mapping enhances public health monitoring by providing a near-real-time visual representation of unsafe zones [9]. Health workers can identify high-risk locations, particularly at the household level, and comprehend spatial patterns that are crucial for making informed decisions, which offers a near-real-time perspective of the community. According to Bardiago et al. [2024] and Reyna [2023], systems that integrate Google Maps for geolocating health facilities and patient populations have been found "highly effective" in terms of usability and functionality, significantly enhancing health information management at the barangay level [3, 10]. This finding supports the LMLinga project's decision to integrate GIS mapping to locate the household's location and plot it in a digital map. The integration of GIS and automated spot-mapping will enhance data integrity, but also provide the granularity needed to identify high-risk households with coordinate-based precision [16]. By moving away from hand-drawn maps, the proposed system ensures that the household's spatial data remains accurate and actionable for long-term community health planning.

The reviewed studies show that Geographic Information Systems (GIS) and automated spot-mapping technologies are important tools for improving healthcare monitoring and data management in local communities. These studies are relevant to the LMLinga project because they support the use of digital mapping to replace the traditional hand-drawn maps currently used in Barangay La Medalla. Through the integration of GIS and automated spot-mapping, the proposed system can help Barangay Health Workers identify high-risk households more accurately, monitor community health conditions in real time, and update household information more efficiently. The findings also support the effectiveness of integrating mapping technologies such as Google Maps and web-based GIS in improving usability and healthcare decision-making at the barangay level. Overall, the reviewed literature provides a strong foundation for the development of LMLinga's automated spot-mapping feature, helping improve the accuracy, accessibility, and reliability of healthcare monitoring within the community.

2.1.5 Multilingual Chatbots in Health Information Systems

A software application or a computer program designed to engage in conversation using humanlike language is called chatbots [13]. Over time, it evolved from basic text-based input into artificial intelligence (AI), natural language processing (NLP), and machine learning (ML) to learn, adopt and not just to respond for personalization and better interaction. The evolution of chatbots began from a statistical model that progressed through landmark early systems like ELIZA and ALICE, which mimicked dialogue via pattern matching. Another milestone includes Alan Turing's imitation game (Turing Test), influential projects like CALO (Cognitive Assistant that Learns and Organizes), and the transformer-based revolution, enabling modern agents such as ChatGPT and Google Bard [38]. Chatbot evolved from rule-based scripts to sophisticated, context-aware systems capable of handling routine tasks like scheduling or customer service. Today, these tools provide responses, assist users, or automate tasks in various applications across healthcare, education, and beyond.

In healthcare, chatbots act as a digital assistant, provide quick responses, and personalized answers to consumers and patients. It streamlines the administrative process, assesses with assessment, diagnosis, treatment, and empowered health information searches unlike to the traditional web. It offers 24/7 online healthcare support and guide user through a series of questions to clarify queries. This application also enhances the efficiency between professionals and patients by delivering a response with guidelines, giving an instant answer, which reduces navigation time through complex resources [29]. Today, there are platforms that are specialized for clinical support that already exist. In the middle of the pandemic, a chatbot platform enables supportive care without hospital visits via a conversational AI-based application to remote patient connections for rapidly transforming in-person care, which provides healthcare education, information, and guidance [5]. This study supports that chatbots can exist for healthcare, providing health education without visiting a health center and give instants answer available 24/7. Our chatbot feature will not utilize artificial intelligence, just like the related literature. Our non-AI chatbot will provide an information education documents campaign (IEC) that comes from the DOH, assuring that the information they received was legitimate and trusted.

Language barriers occur when using a chatbot because of different culture which shows the need for multilingual support for the health system for better accessibility and personalization. Chatbot like Dr. BOT has an architecture that supports multilingual NLP and fine-tuned neural machine translation (NMT) to preserve medical terminology accuracy across languages [32]. Dr. BOT used machine learning algorithms trained on validated datasets for predicting diseases, including diabetes, heart disease, kidney issues, liver conditions, and breast cancer. This serves as a global tool to improve health outcomes and concerns in underserved communities. Dr. Bot system aligns with our study with multilingual support with the use of NLP for translating users' questions and dataset fine-tuning to deliver accurate responses from Information Education Campaign (IEC) materials for the local health center. Our system adopts this approach especially for our multilingual feature (English, Tagalog, Bicol-Iriga) that ensures accessible health information within the local community.

Locally, the chatbots focus on maternal health, physical, and mental health monitoring, and a Filipino/Bisaya ASR BOT exists. The maternal chatbot is powered by OpenAI's ChatGPT and has been deployed for 10 months. This is used to generate responses to frontline workers' questions. This study emphasizes that general-purpose AI like ChatGPT requires local medical professional review and country-specific training for optimal effectiveness in adhering to regional regulations and protocols. Next, the Filipino ASR Bot focuses on assessing children's physical wellness due to shortages of medical personnel and families' limited resources, often preventing routine check-ups incorporated with automatic speech recognition [30]. It stands as pediatric healthcare uses a custom-collected speech corpus that achieved a low word error rate (4.37%) for Filipino and 7.16% for Bisaya, which shows the effectiveness of monitoring of children's general physical wellness. The multilingual chatbot also exists locally, which is an AI chatbot for mental and physical health monitoring that supports Filipino, Bisaya, and English languages to facilitate frequent checks without full-time nurses [7]. It flags needs for psychological, medical, nutritional, or socio-cultural interventions, promoting early detection in resource-constrained environments. These chatbots in the Philippine context exist with artificial intelligence (AI), including multilingual support for Filipino, English, and Bisaya. In contrast, our system uses only information education campaign (IEC) datasets from the Department of Health (DOH), supporting English, Tagalog, and Bikol-Iriga languages for a specific community.

The existing chatbot on health care, with the use of multilingual literature, highlights the advantage of fine-tuned natural language processing (NLP) for health care. Philippine studies succeed with Filipino/Bisaya but overlook broader dialectal diversity and global tools like Dr. BOT that lack integration of local IEC datasets. This study bridges these gaps by developing a multilingual chatbot supporting English, Tagalog, and Bikol-Iriga, consisted with health center IEC data for education, prevention, and campaigns, which enhanced accessibility and proactive engagement in barangay-level care. By combining high-accuracy NLP with culturally tuned responses, it advances localized health education beyond current prototypes.

The present chatbot for health care reflects the advantages of fine-tuned natural language processing (NLP) in health care through the multilayer use of literature across multiple languages. While Philippine studies using Filipino/Bisaya were successful, there are limitations due to their focus on dialectical diversity and the lack of integration of local IEC datasets into global tools such as Dr. BOT. The gaps in the literature are resulting in the creation of a multilingual chatbot that will support English, Tagalog, and Bikol-Iriga as input languages, and include health center IEC data for use as an educational tool to help promote prevention and increase the ability of barangays to connect with their local health care system. This combination of highly accurate NLP into culturally appropriate responses will allow local health education to be developed and advanced past the current development of any current prototype.

2.1.6 Methodologies and Tools for System Development

2.1.6.1 Methodology

Using a flexible development methodology is important when working on community-based health systems because their requirements can change over time. According to Daraojimba et al. [2024], Agile software development is designed to handle complex projects by dividing them into smaller iterations, which allows developers to improve the system step by step during the development process [15]. It helps reduce the chances of project failure because they encourage regular communication and allows developers to deliver working features early in the development cycle. In addition, the main advantages of Agile is its ability to adapt to changes in user needs while the system is still being developed [39]. For the LMLinga project, this approach allows the researchers to develop the household profiling and mapping features in separate sprints. This makes it easier to test each feature and get feedback from the Barangay Health Workers (BHWs) of Barangay La Medalla before combining everything into the final system.

The studies reviewed indicate that Agile methodology is characterized by its flexibility and ability to respond in an iterative manner to changes in user requirements and its provision for continued engagement with stakeholders throughout the software development process. Consideration of incremental software development, regularly receiving feedback, and providing early delivery of software components all lead to the production of dependable and user-centered software applications. Therefore, these findings are significant as they supply appropriate guidance in the development of the LMLinga system for managing developing requirements in health care and incorporating the needs of Barangay Health Workers into their development on a continuous basis.

2.1.6.2 Tools Used

Modern web applications also need structured frameworks to keep the code organized and ensure that the system performs well in the long term. Using structured frameworks helps developers build institutional systems more efficiently because it provides a clear structure for the system architecture. With the help of structured frameworks, the LMLinga will be developed in an organized and stable way. It helps reduce coding errors and makes the development process faster, which is helpful in meeting the needs of the health center. Subecz (2021) also states that frameworks like Laravel use the Model-View-Controller (MVC) pattern, which separates the business logic from the user interface, making the system easier to manage and maintain [34]. By using the MVC pattern, the system becomes easier to manage because the user interface is separated from the main system logic. This allows developers to update or change the interface of the health center's system without affecting or damaging the resident health data stored in the system. Furthermore, Fababeir et al. [2024] mention that frameworks can improve developer productivity and system security because they automate common tasks such as database routing and data validation [40]. By applying these concepts, the LMLinga system ensures that the backend responsible for handling resident records and mapping data in Barangay La Medalla remains organized, secure, and easier to expand in the future.

Interactive mapping is also an important feature for improving the manual spot-mapping process currently used in many local health centers. According to Marie [2026], Leaflet.js is a lightweight and open-source JavaScript library that allows developers to create interactive maps that can run efficiently in web browsers [25]. Because Leaflet.js is lightweight, the automated spot-map can load quickly even on the basic mobile devices that Barangay Health Workers usually use while working in the field. Leaflet has a modular structure, which means developers can include only the components they need, helping improve page loading speed and making it more suitable for mobile devices [22]. This modular setup keeps the LMLinga system running smoothly, even in areas of the barangay with slow mobile internet. In LMLinga, this feature lets the automated spot-map load quickly on the mobile devices used by Barangay Health Workers. The system takes the latitude and longitude from the database and plots them on the map automatically, so health workers can easily locate the households' locations in real time.

The integration of artificial intelligence in community health services also requires methods that ensure both accuracy and privacy of information. According to Koutsiaki et al. [2025], Retrieval-Augmented Generation (RAG) allows chatbots or “Talkbots” to provide more reliable responses by retrieving information from trusted sources before generating answers [23]. RAG technology helps the LMLinga chatbot give more reliable health information because it gets its answers from official DOH medical documents instead of generating responses on its own. Neha et al. [2025] also explain that RAG helps prevent AI hallucinations because the system first retrieves verified information before producing a response [28]. By reducing AI hallucinations, the system can provide more accurate health information. This helps the community trust the system because residents get accurate and safe health advice in their own dialect. To keep sensitive data even more secure, Matotek et al. [2025] recommend using local runtimes like Ollama, which let the language models run on a private server instead of sending information to the cloud [26]. Using a local runtime like Ollama helps keep all resident interactions with the chatbot private because the data stays within a local server instead of being sent to public cloud servers. This also helps the system follow the requirements of the Data Privacy Act. In the LMLinga system, these technologies allow the chatbot to retrieve verified information from the Department of Health (DOH) guidelines and community health records. This enables the system to provide accurate health-related responses to residents in the Bikol-Iriga dialect while still following the requirements of the Data Privacy Act.

A reliable database is also necessary for keeping health records organized and accessible. MySQL continues to be widely used in web systems because it is stable and capable of handling large amounts of relational data [40]. MySQL offers a strong and reliable way to store data, making it possible to manage the increasing health records of hundreds of households in Barangay La Medalla for many years. Subecz [2021] also notes that combining a relational database with a framework improves data validation and overall security [34]. In the LMLinga system, MySQL stores household profiles and individual health records in connected tables. This setup keeps the information organized and reduces the chance of duplicate records. It also helps the health center quickly generate reports whenever they are needed.

The literature supports the necessity of selecting the appropriate development tool and technology for creating efficient, scalable, secure and maintainable systems. Examples of development tools include frameworks, such as Laravel, mapping libraries like Leaflet.js, systems for retrieving information such as Retrieval-Augmented Generation (RAG), local runtimes for Artificial Intelligence (AI) such as Ollama, and relational databases like MySQL will all add value to building out reliable, structured information systems. The development tools discussed above are important to the study being proposed, as they will provide the necessary technological infrastructure to implement household profiling; the ability to map out spots in an automated manner; the ability to build a chatbot with multiple languages; and to manage health records in a secure manner through the LMLinga system.

2.1.6.3 Evaluation

The researchers will use Black Box Testing to check if the system works based on its functional requirements. According to Ayuningtyas et al. [2023], this type of testing focuses on inputs and outputs without looking at the internal code [6]. This helps ensure that Barangay Health Workers (BHWs) can easily use the profiling forms and mapping tools. It is also useful for User Acceptance Testing (UAT) and Formative Usability Testing because it gathers feedback from actual users, especially those who are not technical. Alshamaa et al. [2025] explain that this method can detect interface errors and database issues, helping confirm that the system is reliable [4]. It also supports Integration Testing by checking if the Laravel backend, MySQL database, and Leaflet.js mapping system work properly when combined.

Ayuningtyas et al. [2023] states that verification is important to make sure the system functions correctly and has fewer errors before deployment [6]. This is important in Regression Testing because when updates are added, such as improving the chatbot's Bikol-Iriga responses, the existing features like health records should still work properly. The system needs regular testing to keep its performance consistent as new features are introduced. Nafez Jalal et al. [2026] explains that evaluating the system from the user's perspective helps show how it works in real situations [27]. This is also part of Security Testing to make sure unauthorized users cannot access sensitive information. The process helps find possible issues and keeps the system safe and easy to use for health workers in Barangay La Medalla.

The reviewed literature establishes the importance of comprehensive evaluation methods in verifying software quality, reliability, usability, and security. Testing types such as Black Box Testing, User Acceptance Testing, Integration Testing, Regression Testing, and Security Testing support defect detection, requirements validation, and confirmation that the system is operating well in a typical environment. These evaluation methods provide a structured process to evaluate the performance/functionality and user acceptance of the LMLinga system prior to its going live at the La Medalla Barangay Health Center.

2.1.7 Software Testing and Quality Assurance in Health Information Systems

Systematic testing is a critical phase in the development of health information systems (HIS) to ensure both data reliability and patient safety. Because these systems manage sensitive medical records, software failures can lead to inaccurate community health monitoring or severe data privacy breaches [41]. Quality assurance (QA) in this context involves a structured, multi-layered testing approach to confirm that the system meets all user requirements and operates dependably within a healthcare environment [42]. This process is vital for preventing data loss and ensuring that Information Education Campaign (IEC) materials remain accurate and accessible. Modern healthcare applications require specific validation for data integrity and clinical accuracy to avoid the spread of health-related misinformation [37]. For the LMLinga system, rigorous testing validates that the integration of Leaflet.js for mapping and Ollama for the chatbot does not cause system crashes or performance degradation.

Aligned with the Agile methodology, the study utilizes formative usability testing as an iterative process to gather continuous feedback from Barangay Health Workers (BHWs). Recent research indicates that involving local health workers in the testing phase significantly improves the adoption and ease of use of digital health tools in primary care settings [2]. This ensures that the user interface is intuitive for field operations and that the multilingual chatbot effectively processes and responds to queries in English, Tagalog, and Bikol-Iriga.

Furthermore, integration and security testing are performed to confirm that the database, web frontend, and other components work together seamlessly. Security testing is particularly critical under the Data Privacy Act of 2012, which requires Philippine health systems to implement technical safeguards like Role-Based Access Control (RBAC) and OTP verification to protect resident records from unauthorized access. To maintain system quality throughout the development lifecycle, regression testing is conducted to ensure that new updates do not break existing features, such as the offline data synchronization mechanism [24]. This ensures that the system remains stable even as changes are made based on user feedback. The testing cycle concludes with User Acceptance Testing (UAT), where the BHWs and the head midwife perform a final evaluation. This stage verifies that the digital system effectively replaces manual, paper-based workflows and meets the operational standards required by the Barangay Health Center of La Medalla.

The reviewed studies highlight the importance of proper software testing and quality assurance in developing reliable and secure health information systems. These studies are relevant to the LMLinga project because the system handles sensitive resident records and supports the daily operations of Barangay Health Workers in Barangay La Medalla. Through continuous testing, usability evaluation, and security validation, the researchers can ensure that the system functions properly and remains easy to use for healthcare workers. The findings also support the need to maintain system stability and accuracy as new features are added, helping LMLinga provide dependable household profiling, automated spot-mapping, and multilingual chatbot services for the community.

2.1.8 System Deployment and Effectiveness Evaluation in Healthcare

The deployment of Health Information Systems (HIS) in community settings represents the stage where a healthcare system begins actual operation after its initial development. Epizitone et al. [2023] described system deployment as a strategic effort to modernize healthcare processes by creating an integrated digital ecosystem that brings different healthcare functions into one system [17]. This concept is similar to the objective of the LMLinga, which aims to replace fragmented manual records with a unified digital platform for Barangay La Medalla. Sucaldito et al. [2025] found that digital health adoption in the Philippines continues to increase, but local health centers still experience challenges during system deployment due to unstable internet access and varying levels of digital literacy among staff [35]. This situation highlights the need for a system that can still function even with limited connectivity, allowing Barangay Health Workers (BHWs) to record important health data without relying on constant high-speed internet.

The study of Gonzaga Ballaran et al. [2023] evaluated a web-based barangay profiling system and reported high scores in functional suitability and security performance. The study explained that digital systems reduce data duplication and minimize administrative errors [19]. These results provide a reference for the development of LMLinga, indicating that automated household profiling can improve data accuracy compared to manual record-keeping. Abbasian et al. [2024] explained that the effectiveness of a system depends on the reliability and trustworthiness of the information it provides, especially when users rely on it as a main source of information [1]. This highlights the importance of using verified health center data in LMLinga so that residents receive accurate and reliable health information.

System components that do not rely on artificial intelligence can still be effective because they maintain logical consistency and accurate processing of information. Singh et al. [2023] showed that rule-based multilingual interfaces improve user engagement by providing clear and predictable responses, which help address language barriers in local communities [33]. This finding supports the LMLinga project's decision to use a rule-based chatbot, as it provides consistent health answers in Bikol-Iriga that are easier for residents to understand and trust. Digital spot-mapping offers better spatial analysis than traditional paper maps because it allows users to observe community health patterns more clearly [21]. This capability allows the La Medalla health center to identify and visualize possible high-risk areas more effectively than hand-drawn maps. Tanwar et al. [2023] also emphasized that the success of healthcare communication tools depends on the accuracy of the responses they provide. Pre-programmed logic helps ensure that the information given to users remains accurate and reliable, which is important when sharing health-related information [36]. This approach allows LMLinga to provide verified and stable responses to health inquiries while reducing the risk of incorrect medical information.

The literature explicitly focusing on system deployment strategies and effectiveness evaluations highlights the core parameters needed to successfully launch and measure digital health innovations in rural communities, directly guiding the evaluation phase of the LMLinga project. Previous empirical studies indicate that while digital ecosystems successfully modernize primary care, deployment effectiveness in Philippine barangays is routinely constrained by unstable internet connectivity and low digital literacy among field workers. This evidence directly justifies LMLinga’s architectural integration of offline-first capabilities, utilizing localized storage and synchronization tools to ensure uninterrupted system operations during field-level profiling in Barangay La Medalla. Furthermore, empirical validation from similar barangay profiling systems demonstrates that establishing logical data consistency, role-based access control (RBAC), and automated reporting modules maximizes data security, minimizes computational latency, and fosters community trust. Ultimately, these established frameworks provide the researchers with structured evaluation methodologies, such as utilizing system response times, data integrity metrics, and user feedback loops, to objectively confirm that the deployed LMLinga platform effectively minimizes human error, decreases reporting delays, and enhances primary healthcare delivery.

2.2 Synthesis of the State-of-the-Art

The reviewed literature and systems show that the digital transformation of primary healthcare is supported by three major technologies: web-based household profiling, Geographic Information Systems (GIS) for automated mapping, and multilingual chatbots. Dahilan et al. [2025] and Sucaldito et al. [2025] reported that Barangay Health Centers (BHCs) experience operational challenges when using manual paper-based records, which often result in slow reporting and inaccurate data [14, 35]. Elepaño [2025] proves that a more effective way to address these issues is through the use of the Extreme Programming (XP) agile method that allows developers to design systems directly with healthcare providers instead of using a traditional plan-based approach to project management [16]. This methodology is based on user stories and acceptance criteria that help developers collaborate with Barangay Health Workers to break projects into manageable pieces to develop critical features while ensuring the protection of sensitive information. Web-based barangay profiling systems developed by Gonzaga Ballaran et al. [2023] and Jacobe et al. [2021] demonstrate how digital household profiling can improve data management by organizing records and supporting automated reporting [19, 20]. LMLinga follows a similar approach by using a digital household profiling framework to improve the accessibility and management of resident records. These systems support better data governance and help barangay offices perform evidence-based planning. LMLinga expands these existing systems by combining record digitization and document management with automated spot-mapping and multilingual chatbot functions in a single platform that supports administrative monitoring, health tracking, and health education.

Balba et al. [2023], Bardiago et al. [2024], and Jacobe et al. [2021] demonstrated that GIS and geo-mapping technologies can display health trends and identify areas with higher health risks [9, 10, 20]. The integration of mapping tools such as Google Maps with spatial analysis systems allows better visualization of community health conditions and supports decision-making in health monitoring. LMLinga applies these concepts through its automated spot-mapping feature, which displays household data on an interactive digital map. Existing systems use GIS to support community health analysis through visual data representation and spatial analysis. Many of these systems operate at the municipal or city levels. LMLinga focuses on household-level mapping within Barangay La Medalla. A key technical feature of LMLinga is its direct and real-time synchronization between the profiling database and the map interface. Unlike other systems that need manual updates to change the map, LMLinga automatically updates the spot-map whenever a Barangay Health Worker (BHW) saves a new household record. This means the information on the map is updated right away without needing extra steps.

Babu et al. [2024], Ni et al. [2023], and Rahul et al. [2026] showed that AI-powered chatbots in healthcare can improve patient communication, provide easier access to information, and reduce administrative tasks [8, 29, 32]. This trend reflects the increasing use of automated systems for patient engagement. The LMLinga system plans to adopt this approach to help reduce the heavy documentation workload of Barangay Health Workers (BHWs). The studies of Azcarraga et al. [2024], Pascual et al. [2023], and Singh et al. [2023] on the Filipino-Bisaya system and the Dr. BOT multilingual healthcare chatbot highlight the importance of language inclusion and contextual adaptation in system development [7, 30, 33]. LMLinga aims to improve accessibility through multilingual support, including the use of the Bikol-Iriga language. However, unlike several generative AI systems that rely on broad medical datasets, LMLinga's chatbot is designed to incorporate localized Information Education Campaign (IEC) materials specific to the Barangay Health Center of La Medalla. This design addresses the concerns raised by Chen et al. [2025] and Abbasian et al. [2024] about the need for local validation and reliability in AI-based medical response systems [1, 11]. To make sure the system gives reliable information, LMLinga uses a Retrieval-Augmented Generation (RAG) approach that is powered by Ollama. With this setup, the chatbot generates answers based on verified documents provided by the Department of Health (DOH). Because of this, the responses stay rule-based and accurate, which helps prevent the “hallucinations” or incorrect information that can sometimes happen with general-purpose AI models.

The LMLinga system is evaluated using several testing methods, including formative usability testing with Barangay Health Workers (BHWs), user acceptance testing, system integration testing, regression testing, and security testing. Security features include Role-Based Access Control (RBAC) and One-Time Password (OTP) verification. These practices follow common standards used in health information systems. The development of LMLinga uses the Agile Software Development Methodology. This approach allows the system to be improved step by step through continuous updates. It also makes it easier for the developers to make changes based on the feedback and suggestions given by the Barangay Health Workers (BHWs). The system is also designed for barangay environments with limited resources. Offline synchronization features are included to allow the system to continue functioning even with unstable internet connectivity, which is a common challenge in Philippine primary healthcare settings.

The state-of-the-art shows that existing systems provide separate solutions for household profiling, GIS-based monitoring, and multilingual chatbot support. These systems do not yet offer an integrated solution that fully meets the needs of barangay healthcare operations. LMLinga addresses this gap by combining these three functions into one interoperable platform. The system includes web-based household profiling, automated GIS spot-mapping, and a locally contextualized multilingual chatbot. This design allows the system to support multiple health operations within a single digital platform.

2.3 Gap Bridged by the Study

While existing health information systems have addressed basic record-keeping, LMLinga bridges several critical gaps identified in current manual and digital workflows between 2021 and 2026. One major gap is the reliance on hand-drawn spot maps in local health centers, which are often difficult to update and prone to significant human error. This study addresses this limitation by using Leaflet.js to automate digital spot-mapping, providing a more accurate and real-time visualization of community health trends compared to traditional manual methods. By integrating spatial data directly with household profiles, the system allows health workers to identify clusters of health issues more efficiently.

Another significant gap exists in the linguistic accessibility of health technology, as most global health chatbots are limited to major languages like English or Tagalog. This project bridges this gap by integrating a chatbot that specifically supports the Bikol-Iriga dialect. By utilizing Retrieval-Augmented Generation (RAG), the study ensures that health information is not only linguistically accessible to the residents of La Medalla but is also pulled from verified Department of Health (DOH) datasets. This ensures that the automated assistance provided is both culturally relevant and medically accurate.

Furthermore, the study addresses the digital divide caused by unstable internet connectivity in rural Philippine communities. Most web-based systems require a constant connection, which creates a gap in service during house-to-house profiling in remote areas. LMLinga bridges this through an offline-first approach, using local storage and background synchronization to ensure that BHWs can continue their work without data loss. Finally, the study provides a secure and verified method for resident data access. While many systems lack a secure portal for residents, this project implements RBAC and Head Representative verification to create a secure link between the health center's records and the residents' right to access their data, adhering to modern standards for patient-centered data access.

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CHAPTER 3
TECHNICAL BACKGROUND

This chapter presents the methodological and technical foundation of the proposed system, detailing the strategies and tools required for system development. It explains how the researchers will develop the system and how they will ensure that the system works effectively based on the client's needs.

3.1 Research Design and Methods

The study will employ a mixed-method research design that combines quantitative, qualitative, and developmental approaches to gain a more comprehensive understanding of the research problem rather than a method alone. This approach capitalizes on the strengths of both qualitative and quantitative research, providing a broader range of perspectives on complex subjects. Mixed-method research is commonly used for purposes such as triangulation, complementarity, development, initiation, and expansion [6]. By combining numerical data and detailed responses, the researchers can improve the accuracy and quality of the analysis and conclusions. This also helps increase the validity and reliability of the findings. Mixed-method research is considered useful for modern researchers because it is flexible and can adapt to more complex research needs [13]. This design is suitable for LMLinga because it specifically bridges the achievement of the project's objectives.

The qualitative component entails collecting and analyzing user stories, work processes, and technological criteria for how users manage health information through household data profiling, record management, report generation, and spot mapping in relation to health information availability at the health center. Qualitative methods include semi-structured interviews and observational data collection in order to comprehend how Barangay Health Workers (BHWs) conduct household profiling, manage patient records, and perform manual spot mapping effectively. The methods used to collect qualitative data will provide BHWs with comprehensive requirements that are centered on the needs of users, while identifying various functionality and performance criteria that will guide researchers in developing a centralized health information system. Qualitative data will be grouped thematically to identify similarities and differences in how BHWs perform similar functions and create common user experiences, as well as functional specifications and requirements for the LMLinga application system based on XP methodology for determining the application's technical specifications and acceptance criteria.

The testing phase will apply an experimental quantitative research design using black box test cases, OWASP ZAP security scanning, and Apache JMeter load testing to assess the system's functionality, security, and performance. These testing methods will be used to verify that the system meets the required standards in terms of accuracy, reliability, and responsiveness. Lastly, the evaluation phase of the system will use an evaluative mixed methods research design through a 5-point Likert scale survey answered by BHWs and IT experts, along with direct observation of BHW system usage to gather both numerical and descriptive data.

3.1.1 Software Development Methodology

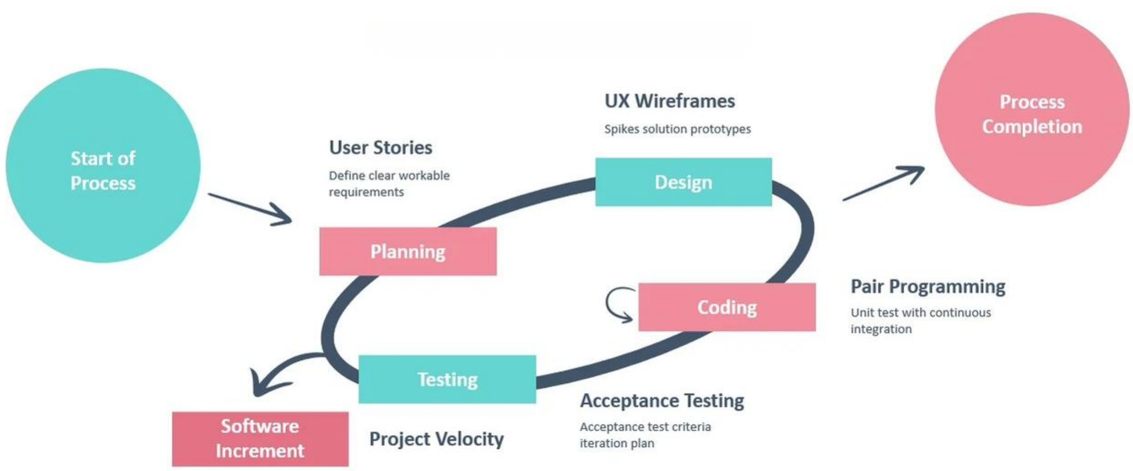

Figure 1. **Extreme Programming under Agile Methodology [15]**

Planning. The first step of the development process is to identify how the LMLinga system will be implemented. Therefore, the researchers will work with the Barangay Health Workers

(BHW) to gather user stories and ask them about their workflows and what criteria they need to accept new technology as an acceptable average system. The researchers will then gather detailed information regarding the current processes used in profiling a household, managing health records, generating reports, performing spot mapping, and providing healthcare information services through both semi-structured interviews and observational activities within the Barangay Health Center.

Design. In this phase, it will translate the identified requirements into a structured representation of the system. The team will focus on creating visual models such as interface layout, process flows, system architecture, and structure on how the system will function. This stage will serve as a blueprint that guides the development process by defining the system's structure and user interaction.

Coding. In this phase, it is a building stage, which involves the actual development of the system by transforming the designed plan into software using chosen programming languages and technologies. The development process includes writing source code, performing testing, reviewing code, integrating modules, and debugging issues to create operational software. The iterative approach can also begin in this stage whenever the client has a suggestion or modification about the system. Afterwards, designing, coding, and testing after performing the changes based on client feedback.

Testing. In this phase, it ensures that the developed system functions correctly, securely, and in accordance with the specified requirements. Once the current build iteration has been coded and implemented, the next step is a variety of testing, including user acceptance testing, formative usability testing, integration, regression, and security testing. The mentioned testing methods help to ensure that the system is reliable, user-friendly, and secure before deployment

3.1.2 Data Sources

The primary and secondary information that will be used to accomplish the study goals are provided in this section. Specifically, this information will be used to collect the user stories, evaluate technical acceptance criteria, and assess the usability and functionality of the proposed LMLinga system. The data collected from users and through evaluations will be utilized as part of the Extreme Programming (XP) framework to determine whether the proposed centralized healthcare information system is effective, usable, and operationally viable for the Barangay La Medalla Health Center.

Sampling Technique

The research will utilize appropriate types of probability sampling as well as different non-probability types of sampling that are tied to the research objective and identified respondent groups. The selection procedures for the sample will support the collection of user stories, operational requirements, technical acceptance criteria, and evaluation feedback necessary for the development and evaluation of the proposed LMLinga system.

The total enumeration sampling technique will be used for conducting research with the Barangay Health Workers (BHWs). The number of BHWs working in Barangay La Medalla Health Center is relatively small; therefore, they will all be used as respondents for this research. In this way, there is complete participation of all personnel who are directly involved in the household profiling, spot mapping, report generation, and healthcare record management [8]. Their responses will provide user stories, operational workflows, and technical acceptance criteria necessary for the XP framework based on the collected data for the study.

For resident respondents, stratified random sampling will be used to ensure proportional representation of the five different zones of the barangay. In order to determine the resident respondent sample size for this study, Slovin's Formula with a 95% confidence level and 7% margin of error will be utilized. While the usual use of 5% margin of error has been well documented for population-based surveys that require higher degrees of statistical accuracy, a margin of error should be chosen according to the goals and intended use for study results, not according to a fixed standard [11]. The intention for this group of respondents is to evaluate the usability of specific resident-related functions associated with the proposed system and not to reflect the opinions of all barangay residents or constituents. There has been research demonstrating that adequate sampling depends upon statistical audits, data quality, feasibility, and the relevance of the respondents, as well as providing the foundation for meaningful evaluation results [5]. Therefore, using 7% for a margin of error enables the researcher to adequately determine the sample size required for adequate usability testing, as well as manageability associated with obtaining the data. The obtained sample size shall be distributed proportionally across the five identified barangay zones to ensure residents have equal representation. Stratified Random Sampling is suitable for this study since it will provide representative feedback from residents about the accessibility and usability of both the multilingual chatbot, which provides healthcare information through the LMLinga interface, and the overall effectiveness of the system's healthcare information functionality [14].

Slovin's Formula

$$n = \frac{N}{1 + Ne^2}$$

where:

n = sample size

N = total population

e = margin of error

$$n = \frac{2,878}{1 + (2,878)(0.07)^2}$$

$$n = \frac{2,878}{1 + (2,878)(0.0049)}$$

$$n = \frac{2,878}{15.1022}$$

$$n = \frac{2,878}{15.1022}$$

$$n = 191$$

Proportional Allocation per Zone

$$n_h = \frac{N_h}{N} \times n$$

where:

n_h = sample for the zone

N_h = population of the zone

N = total population

n = total sample size

Table 1: Proportional Allocation per Zone

ZONE	TOTAL POPULATION	COMPUTED SAMPLE
1	605	40
2	615	41
3	553	36
4	755	50
5	350	23
TOTAL	**2,878**	**190**

As shown in Table 1, the resident respondents were distributed proportionally to the populations of the five zones of Barangay La Medalla. Based on a total sample size of 190 residents determined with a 7% margin of error and using Slovin's formula, the researchers allocated respondents to each zone proportionally so that each zone would be adequately represented according to their share of the total population. The resulting distribution of respondents included 50 residents from Zone 4, which has the highest number of residents, given that the zone has the greatest population (755), 41 residents from Zone 2 with a 615 total population, 40 residents from Zone 1 with a 605 total population, 36 residents from Zone 3 with a 553 total population, and 23 residents from Zone 5 with a 350 total population. In this way, the researchers ensured that residents from all parts of Barangay La Medalla had a fair chance of being included and, thus, that the collected data would be useful for evaluating all aspects of usability regarding the resident-related functions of the proposed system.

For IT Experts, the study will use purposive sampling, in which participants are selected based on specific characteristics and criteria. The purposive sampling approach will allow for the identification of an appropriate participant sample size to evaluate the system's technical functionality, usability, security, reliability, and overall software quality. These evaluations by the selected IT experts will be instrumental in validating the technical acceptance criteria for the developed software and will demonstrate that the resultant product is in alignment with applicable software engineering practices and standards. This method is most effective for qualitative and quantitative research when the goal is to gather information from experts within a specific cultural domain [14].

Respondents

This section identifies the individuals who will participate in the study and will provide relevant data for the evaluation of the proposed system. The selected respondents are directly involved in the daily operations in the health center, enabling the researcher to gather direct user-based feedback and operational insights. Table 2 presents the distribution of respondents included in the study based on their respective roles.

Table 2: Respondents

RESPONDENTS	SAMPLE	PERCENTAGE
Barangay Health Workers	10	4.88%
Residents	190	92.68%
IT Experts	5	2.44%
TOTAL	**205**	**100%**

Table 2 displays the sample's structure for the study, which is made up of BHWs (Barangay Health Workers), residents, and IT professionals. There are ten (10) BHWs in the sample that will perform household profiling, record-keeping, report generation, and spot mapping at the Barangay Health Center; therefore, they were recruited using a total enumeration method. The purpose of having BHWs participate in the study was to assist in identifying the system requirements, validating the system, and evaluating the system's functionality. The residents in the study come from a sample of one hundred ninety (190) residents selected using a stratified random sampling method that is distributed proportionately across the five (5) zones of Barangay La Medalla. These residents will evaluate various functions associated with their use of the system, such as logging in, OTP verification, entering requests associated with their household, accessing chatbots, and viewing records. Finally, there are five (5) IT experts who will provide a technical evaluation of the system and give their expert assessment of the quality and feasibility of the system.

Secondary Sources

The secondary data sources that will support the establishment of the validity and the refinement of LMLinga include healthcare records and existing operational documentation. According to Elepaño et al. [2025], these records will provide the actual healthcare data, workflow structures, and reporting requirements for primary healthcare systems [2]. These records are essential in ensuring that LMLinga is created to reflect the operational realities of the barangay health workers and provide them with the ability to manage records accurately.

The included secondary sources for this study will include pre-existing forms used in profiling households, residents' records, accomplishment reports, spot-mapping documents, demographic information, monitoring sheets, and any other records related to health that are kept at the Barangay Health Center. These documents will be utilized as references to determine the needed data fields, report formats, and operational procedures that must be programmed into the proposed system. In support, Epizitone et al. [2023] state that organized and reliable records of patient care are essential for improving data access, reporting, and service [3].

Moreover, the analysis of reports and records will support the development of specific modules for the system, such as the digital household profiling, automated report generation, GIS-based spot mapping, and multilingual chatbot. According to Gonzaga Ballaran et al. [2023], existing barangay records and profiling documents serve as an important foundation for creating web-based information systems because they provide a framework for organizing resident information and automating local government services [1]. In a similar vein, Sucaldito et al. [2025]

state that the effectiveness of healthcare information systems is greatly enhanced when they are developed in accordance with the actual workflows of healthcare providers in primary care settings [12].

Documented sources gathered will also support the verification and validation of healthcare data collected and used throughout the testing and evaluation of the prototype system. By utilizing actual operational records from the health center as documentary evidence, the study ensures that the proposed system aligns with existing health-care reporting requirements, record management procedures, and service delivery processes in place in the barangay health department environment.

3.1.3 Research Instruments

This section outlines the research instruments that will be used to collect quantitative and qualitative data necessary for the requirement elicitation, development, and system evaluation. These instruments were selected to effectively capture user stories, operational workflows, technical acceptance criteria, and software quality evaluation data under the Extreme Programming (XP) framework.

Interview. Semi-structured interviews will be conducted with Barangay Health Workers to gather user stories, operational workflows, and technical acceptance criteria related to household profiling, record management, report generation, spot mapping, and healthcare information accessibility. This qualitative instrument will help identify the specific gaps in the current process and determine the user requirements needed for the development of the LMLinga system.

Observation. Through observation, researchers can document the actual operations and processes performed by Barangay Health Workers while conducting household profiling, managing healthcare records, preparing reports, and carrying out spot mapping. This observation allows researchers to document the sequential order of operations, data handling processes, and how users of the system interact with one another and the system to be developed. It also provides evidence to support the validation of user stories and acceptance criteria that were collected from the BHWs during the interview process by providing what the actual workflow looks like in a healthcare environment.

Evaluation Survey. Evaluation surveys will be used as the primary quantitative measure for assessing the quality, effectiveness, and user acceptance of the proposed system. Evaluation questionnaires will be based upon selected ISO/IEC 25010 Software Quality Model Characteristics, specifically functional suitability and usability. These quality characteristics have been selected because they are directly aligned with the objectives of this study, which focus on determining if the system performs the required healthcare activities effectively and if the system is easy to use by Barangay Health Workers and residents.

Different groups of respondents will receive their own separate evaluation forms according to their level of participation in the proposed study. The Resident Access Validation Survey will be administered to all resident respondents for evaluation purposes regarding the usability and accessibility of system features, including account login, household request submission, OTP (one-time password) verification, access to the chatbot, and viewing of medical records. The results of this survey will evaluate whether resident-oriented features of the system are easy to understand, easy to access, and easy to operate.

BHWs will give feedback on BHW sign-off forms after they have completed tests on primary operational functions of the system (household profiling, health information management, GIS-based mapping, chatbot involvement, automated reporting, and logging into the system). The LMLinga system will be evaluated by BHWs for Functional Suitability and Usability based on the tasks they accomplished during their operations.

Task Observation Sheets will be produced throughout the course of testing to document how BHWs execute their various tasks, including identifying households, creating GIS-based maps, generating reports, and utilizing the chatbot feature. The task observation sheets will document the amount of time that it takes BHWs to execute each task, the number of errors made while executing each task, the perceived level of difficulty that BHWs experienced when executing each task, and BHWs' input as to how the LMLinga System has provided support in executing BHWs' responsibilities. The collected observations will assist in the iterative development and validation process related to Extreme Programming (XP).

An evaluation survey will also be administered to a sample group of designated IT professionals to gain additional technical validation of the system under review. Technical validation will consist of three components: the ability of the LMLinga System to meet specifications outlined in this report; the consistency of the LMLinga System; and the overall quality of the LMLinga System with respect to established software engineering guidelines and system development standards.

To measure the technical quality of the proposed system, specific software testing tools, such as OWASP ZAP and Apache JMeter, will also be employed. Security testing will be conducted using OWASP ZAP to identify potential web application vulnerabilities and associated security risks for the proposed system. The performance evaluation and load testing of the proposed system will be conducted with Apache JMeter to determine the responsiveness, operational stability, and load capacity of the proposed system under simulated user activity.

3.1.4 Data Gathering Procedure

Before beginning the research and system development phases, the researchers should obtain the necessary approvals and coordinate with the Barangay Health Center of La Medalla. After obtaining approval to conduct research, the researchers will conduct semi-structured interviews with Barangay Health Workers (BHWs) and other healthcare personnel to collect information on household profiling, healthcare record management, report generation, spot mapping, and healthcare information access. The information obtained will focus on the users in order to identify functional and non-functional requirements for the proposed LMLinga system.

After completing the interview process, researchers will perform direct observation of the operational practices and workflows as they are performed by the BHWs to conduct household profiling, spot mapping, healthcare record management, and the generation of reports. This will allow researchers to record the sequence of operational activities being performed, identify existing workflow structures, and validate the user stories and acceptance criteria developed during the interview portion of the study.

Extreme Programming (XP) under Agile methodology will be employed for the development of LMLinga. System development will take place through iterative sprints of development, where the design, development, testing, and refinement of each module will occur through iterative cycles and based on feedback from stakeholders. After the completion of each sprint cycle, the completed functionality will be presented to the Barangay Health Workers for validation and to get feedback. This will make sure that the features provided can fulfill the stakeholders' (Barangay Health Workers) expectations of their functional and technical data. Ongoing collaboration among stakeholders will continue to support the refinement of system modules, including the digital household profiling, GIS-based spot mapping, multilingual chatbot integration, and automated report generation.

Pilot testing of the evaluation instruments used for the LMLinga program will take place after the development phase has been completed. Researchers will administer evaluation questionnaires to a limited number of respondents and calculate the reliability coefficients for instruments using Cronbach's Alpha. If the reliability coefficients are equal to or greater than .70, this will be taken to show that the evaluation instruments for data collection are reliable and consistent internally.

Pilot testing will be followed by the evaluation and technical testing of the proposed system. Prior to the system being tested, a system demonstration and guided user testing will be conducted with Barangay Health Workers, residents, and IT experts to provide them with an understanding of the LMLinga system features and functions. After the demonstration, the respondents will receive evaluation surveys based on the selected characteristics of the ISO/IEC 25010 Software Quality model (functional suitability and usability). Barangay Health Workers will provide an evaluation of the system's operational functionality and usability, while the residents will evaluate the usability and accessibility of the resident-related LMLinga system features.

During system testing, Task Observation Sheets will be employed to record task completion times, errors, operational difficulties, and user observations that occurred during the performance of household profiling, spot maps, report generation, and use of the chatbot. The recorded observations and evaluation results will be used to support the iterative refinement process dictated by the Extreme Programming (XP) framework.

To evaluate the technical quality of the proposed system, specific software testing tools will be used, including OWASP ZAP and Apache JMeter. OWASP ZAP will be used for security testing to identify vulnerabilities and risks in the web application, while Apache JMeter will be used for performance, efficiency, and load testing to evaluate the system's responsiveness, operational stability, and the ability to handle a specified load during simulated use. The results from the surveys, observations, and technical testing processes will be analyzed and interpreted to determine whether the proposed LMLinga system meets the objectives and software quality criteria set for this study.

3.1.5 Data Analysis Procedure

Thematic Analysis. Qualitative data collected from interviews and observations will be reviewed using thematic analysis. The user stories collected, as well as workflow descriptions and acceptance criteria, will be assigned codes and grouped according to themes in order to define the functional and non-functional system requirements of the proposed LMLinga system. The results from this data will inform both feature refinement and system development using an Extreme Programming (XP) approach.

5-point Likert Scale. A 5-point Likert Scale will be utilized to gather quantitative data on evaluation surveys. Using the ISO/IEC 25010 software quality model, the LMLinga system will be evaluated for Functional Suitability. Functional Suitability is defined as the degree to which a product or product component provides functions that meet stated or implied needs when used under specified conditions at any time or place.

Table 3: Scale for Functionality Evaluation

RANGE	LEVEL OF FUNCTIONALITY	DESCRIPTION
4.20 – 5.00	Excellent	The system completely satisfies the required functions and effectively supports user needs and operational tasks.
3.40 – 4.19	Very Good	The system satisfies most required functions with only minor improvements needed.
2.60 – 3.39	Good	The system performs its intended functions but requires improvements in some areas.
1.80 – 2.59	Fair	The system provides limited support for the required functions and has noticeable operational issues.
1.00 – 1.79	Poor	The system does not adequately satisfy the required functions and user requirements.

As shown in Table 3, the functional suitability of the proposed LMLinga system will be evaluated based on ISO/IEC 25010 software quality model, using a 5-point Likert Scale for the determined level of the proposed system in its function tasks, how well it meets the user's needs, and how well it supports multiple healthcare functions, including household profiling, healthcare record management, spot mapping, chatbot interaction, and report generation. Evaluated using the above methods, the researcher expects to identify the system's strengths and limitations in applicable functional performance capabilities and potential areas for continued development of the system's functional performance.

Table 4: Scale for Usability Evaluation

RANGE	LEVEL OF USABILITY	DESCRIPTION
4.20 – 5.00	Excellent	The system is highly usable, easy to learn, easy to

RANGE	LEVEL OF USABILITY	DESCRIPTION
		navigate, and enables users to complete tasks efficiently and effectively.
3.40 – 4.19	Very Good	The system is generally easy to use and understand, with only minor usability improvements needed.
2.60 – 3.39	Good	The system is usable and supports task completion but has some usability issues that require improvement.
1.80 – 2.59	Fair	The system has noticeable usability issues that affect user experience and task completion.
1.00 – 1.79	Poor	The system is difficult to use and does not adequately support users in completing their tasks.

As shown in Table 4, the Usability of the proposed LMLinga system will be assessed utilizing a 5-point Likert Scale based on the ISO/IEC 25010 software quality model. Usability describes the extent to which an explicit user can accomplish his or her intended tasks effectively, efficiently, and satisfactorily using a specified system [16]. Evaluation of the usability of the proposed system will indicate the level of ease of learning, ease of navigation, ease of understanding, accessibility and overall experience of the user's usability of the proposed system. Based on the results of the analysis of usability, the researchers will be able to determine whether or not the proposed system is usable by prospective end-users, and if so, how effective and/or efficient a tool the system will be in assisting Barangay Health Workers and residents in completing their intended tasks. Also, the researchers will be able to locate areas in need of improvement to enhance the overall user experience of the proposed system.

Weighted Mean. The weighted mean will be used to compute the average ratings from the BHW, IT Expert, and Resident Access surveys. This will convert the responses into grand mean scores. Through this, the researchers can compare the results from all user groups and determine if LMLinga is ready for deployment.

$$\bar{x} = \frac{\Sigma\,(w_i x_i)}{\Sigma w_i}$$

Where:

x = Weighted Mean

$\sum$ = Summation

w = weights

x = values

Cronbach's Alpha. Cronbach's alpha will be used to measure the internal consistency of the BHW, IT Expert, and Resident Access surveys during pilot testing and final survey, with a minimum acceptable value of 0.70. This will help determine if the Likert scale items consistently measure their intended areas, such as usability for BHWs, technical quality for IT experts, and access workflow for residents.

$$a = \frac{K}{K-1}\left[1 - \frac{\Sigma s^2 y}{s^2 x}\right]$$

Where:

K = # of items

$\Sigma s^2 y$ = sum of the item variance

$s^2 x$ = variance of the total score

a = Cronbach's alpha

3.2 Theoretical or Conceptual Background

This section introduces the theoretical and conceptual foundation that guides the design and system development of the proposed system. It details the software development methodology, design pattern algorithms for the technical basis of system design. Figure 1 illustrates the Extreme Programming (XP) under Agile methodology for the development of the proposed system.

3.2.1 Ollama (Large Language Model)

Ollama acts as a local large language model deployment platform with built-in model management and API features that enable developers to build and run conversational chatbots in their own context [7]. The system will use Ollama for the LMLinga chatbot feature as an open-source large language model that supports local hosting while controlling the usage and data privacy. This technology supports the system by generating responses to the resident's query, intelligently processing and understanding user inputs that allows to generate accurate, relevant, and real-time responses. This enhances user engagement, improves communication, and optimizes the overall efficiency and responsiveness of the system.

3.2.2 RAG (Retrieval-Augmented Generation)

Retrieval-Augmented Generation is a natural language processing framework that enhances the accuracy and reliability of large language models by integrating real-time information retrieval [9]. The chatbot will include Retrieval-Augmented Generation (RAG) to ensure that residents receive accurate and legitimate health information. This approach allows AI to pull information directly from the dataset created by the developer from the information education campaign documents for health center and DOH-related health information instead of pulling overall AI knowledge. By grounding the AI's responses in these specific sources, the system prevents hallucinations and focuses on accurate responses. This feature enables residents to receive accurate, quick responses and a trusted digital platform without physically visiting the health center.

3.2.3 Rule-based Filtering

A rule-based serve as a screening layer to establish language parameters and to make sure that the query is relevant before fetching data[4]. To identify the specific topic to be processed in the chatbot, rule-based filtering act as a dual-layered screening component to ensure that the inquiry is relevant and understandable. The first component of the system will use predefined keywords to detect whether the resident's inquiry is in English, Tagalog, or Bikol-Iriga to establish a multilingual chatbot. The second component scans the input for health-related keywords, such as “fever”, “cough,” or specific disease names, to determine whether the query matches the available datasets. Once the keyword is identified, the filter triggers the Retrieval-Augmented Generation (RAG) process to fetch the corresponding information from datasets. This approach prevents the system to response off-topic requests and effectively implements a multilingual chatbot for cultural nuances and breaking language barriers while getting legible information.

3.2.4 Prompt Engineering

The purpose of prompt engineering is to optimize input design and build a restricted behavioral boundary that governs a model’s general flexibility to remain consistent and aligned to a specific service standard [10]. Prompt Engineering is utilized to design specific inputs that guide the language model to act as a professional digital front desk for the health center. Since Ollama is an AI-powered technology, it possesses generative flexibility to respond to any user request; prompt engineering is used to establish strict behavioral boundaries and restrictions. These instructions act as a foundational rule set that prevents the model from turning into unrelated topics. By enhancing the prompts, the developers can effectively govern the language model to produce desired and accurate output. This ensures to remain a helpful, controlled tool that aligns with the service standards of Barangay Health Center.

3.3 System Architecture

This section provides an overview of the overall structure and organization of the proposed system through the 4 + 1 view model of architecture. It is the architectural view of the system from four different perspectives: the functional components of the system; the execution of the processes that make up the system; the organization of the software used to create the system; the environment in which users will deploy the system; and how users will interact with the system, all of which will provide an overall description of how LMLinga will work and provide an efficient, secure & reliable way to implement the system. Figure 2 shows the Logical View of the proposed system.

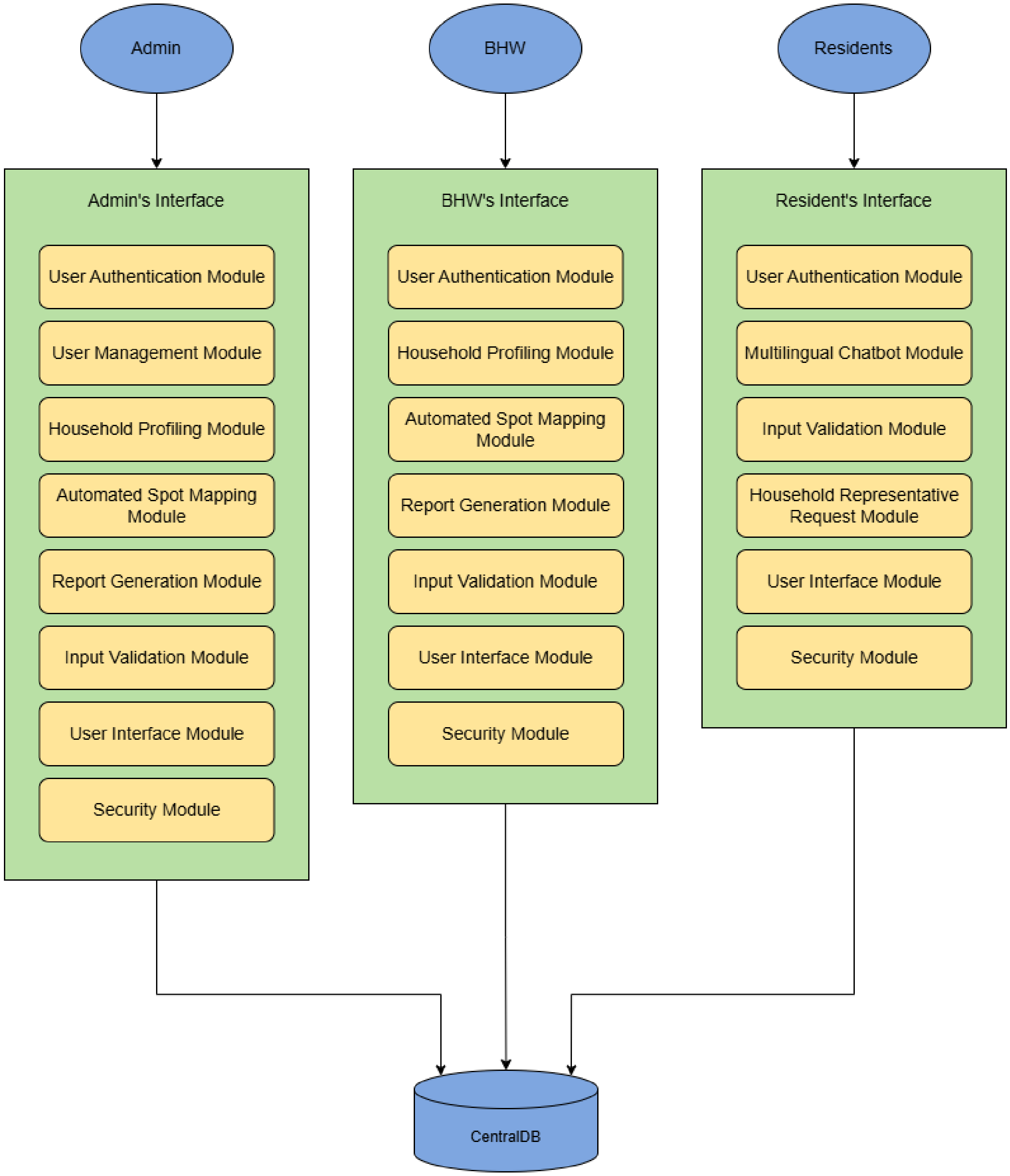

Figure 2. **Logical View**

Figure 2 shows the logical view of the proposed system along with the primary functional components, and how those components relate to each other according to user roles. There are three interfaces within the system: administrator, barangay health worker (BHW), and resident. Each of these interfaces has specific groups of modules that are designed to assist in performing each user's assigned tasks and accessing the system. The administrator interface has modules for user management, household profiling, spot mapping, and report generation. The BHW interface has modules for household profiling, automated spot mapping, and reporting for performing health service operations. The resident interface has modules for user authentication, multilingual chatbot services, and requesting household representatives. Each of these interfaces shares common modules for user authentication, input validation, user interface, and security. All systems and modules will connect to the CentralDB, which provides centralized storage and management of data for the system.

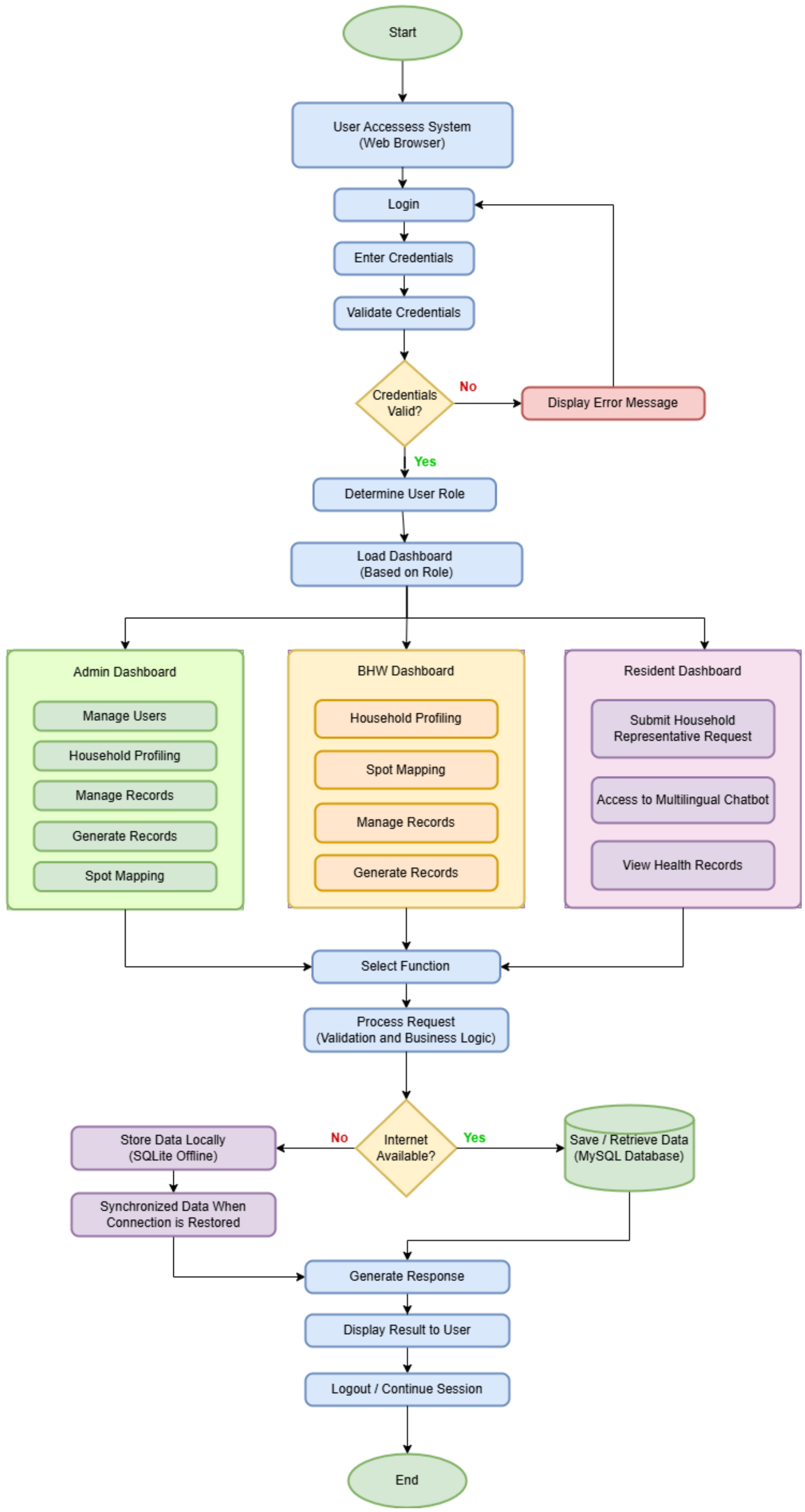

Figure 3. **Process View**

Figure 3 shows the Process View of the proposed system. The system process starts when the user navigates to the system URL in their web browser and logs in with their credentials. If the credentials are not authenticated, an error message will be displayed. If the credentials are authenticated, the system will determine the user's role in the system; based on the user's role in the system, the system will load the role-specific dashboard (Administrator, Barangay Health Worker (BHW), or Resident). After the user has successfully logged into the system, the user can perform those functions available to their assigned role. Every request made to the system will be validated and processed through business logic before the system executes any of the requested functions. The system will also check the availability of the internet when processing data; and when there is an internet connection, the data will either be saved to the MySQL database, retrieved from the MySQL database, or saved to SQLite (for offline use) and will synchronize any data saved to SQLite back to the MySQL database. After processing the request, the system will generate and display the result to the user and allow the user to either continue or log out from the system.

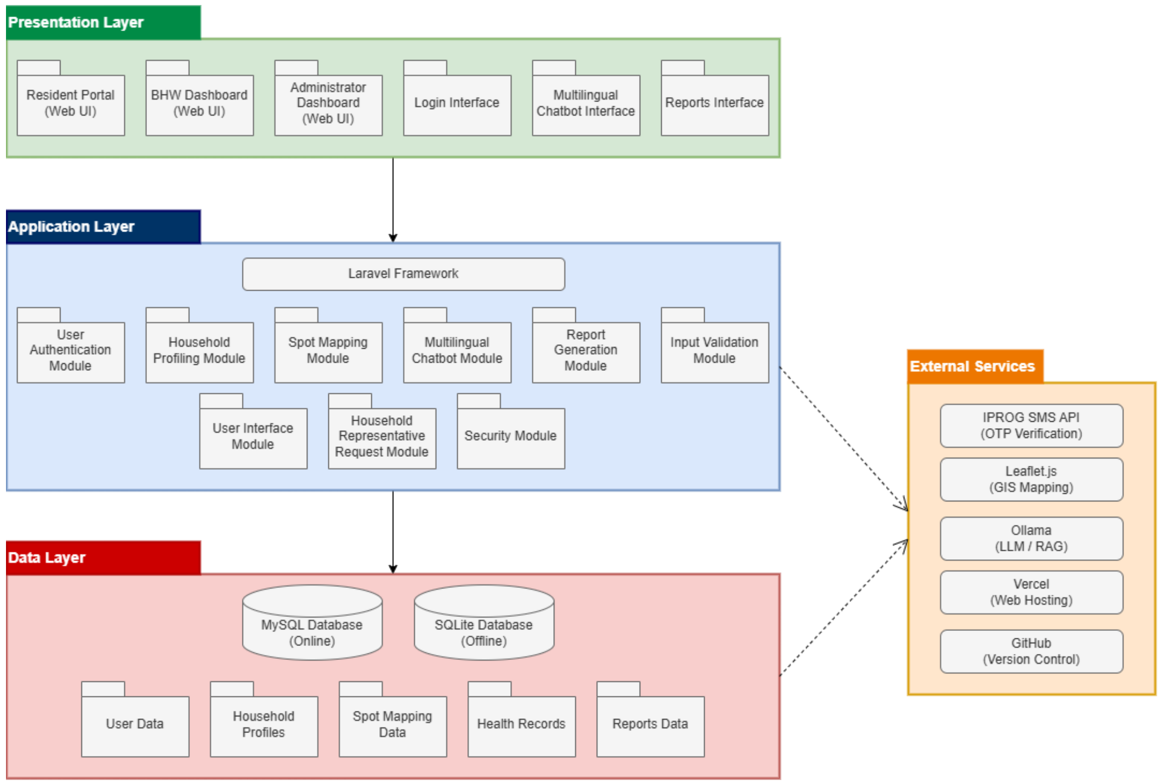

Figure 4. **Development View**

Figure 4 shows the Development View of the proposed LMLinga system, which presents the software organization and implementation structure using a layered architecture. The Presentation Layer contains the user-facing interfaces that allow residents, Barangay Health Workers (BHWs), and administrators to interact with the system through web-based dashboards and modules. The Application Layer contains the Laravel framework and the core system modules responsible for processing requests, implementing business logic, validating inputs, managing household profiling, generating reports, supporting multilingual chatbot services, and maintaining system security. The Data Layer manages the storage and retrieval of information through MySQL for online operations and SQLite for offline data handling, including user data, household profiles, spot mapping data, health records, and reports. Supporting these layers, the External Services provide additional functionalities through integrated technologies such as IPROG SMS API for OTP verification, Leaflet.js for GIS mapping, Ollama for chatbot processing, Vercel for web hosting, and GitHub for version control. Together, these layers organize the system into manageable components to support maintainability, scalability, and efficient system operation.

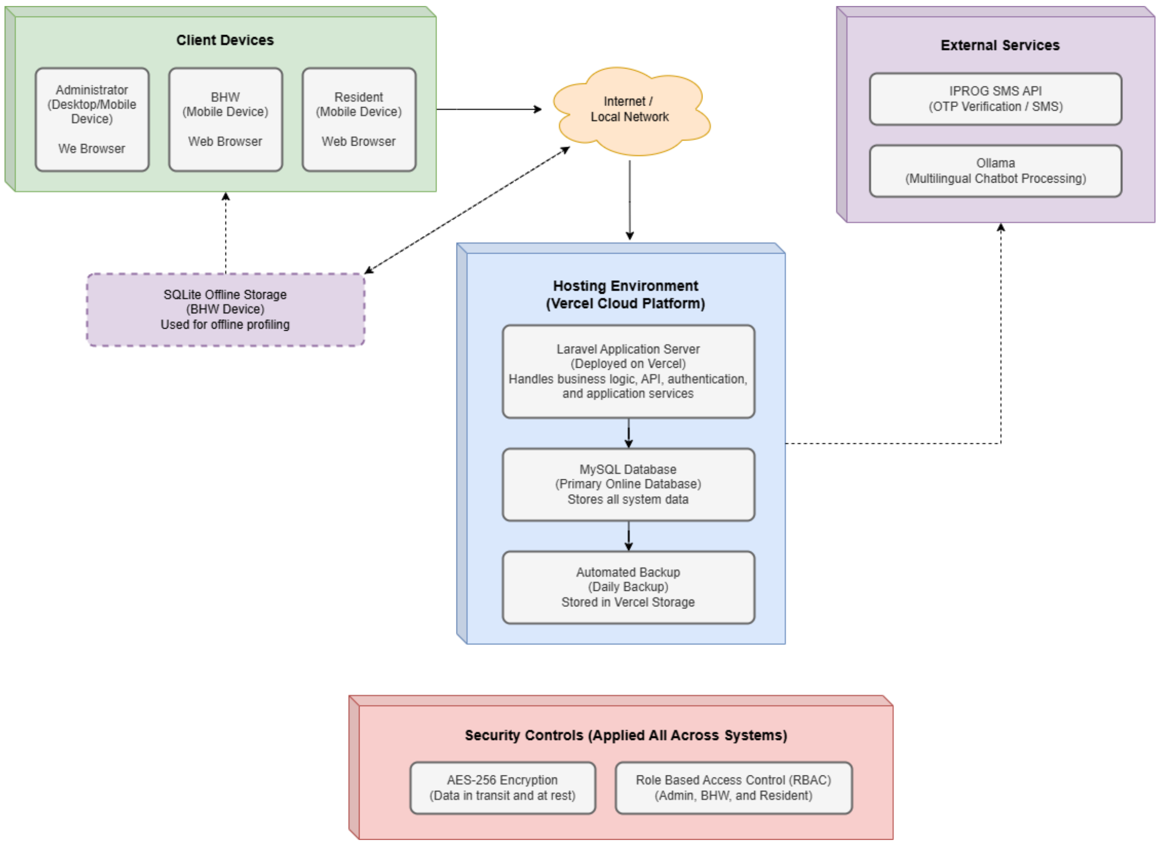

Figure 5. **Physical View**

Figure 5 shows the Physical View of the proposed LMLinga system, which presents the deployment environment and physical arrangement of the system components. Users, including administrators, Barangay Health Workers (BHWs), and residents, access the system through web browsers using desktop or mobile devices connected through the internet or local network. The system is hosted on the Vercel cloud platform, where the Laravel application server manages business logic, authentication, APIs, and application services. Data is stored primarily in the MySQL database for online operations, while SQLite supports offline household profiling activities on BHW devices and synchronizes data once connectivity is restored. External services such as the IPROG SMS API and Ollama are integrated to support OTP verification and multilingual chatbot processing. To maintain secure and reliable system operation, the deployment environment applies AES-256 encryption and role-based access control across all user roles.

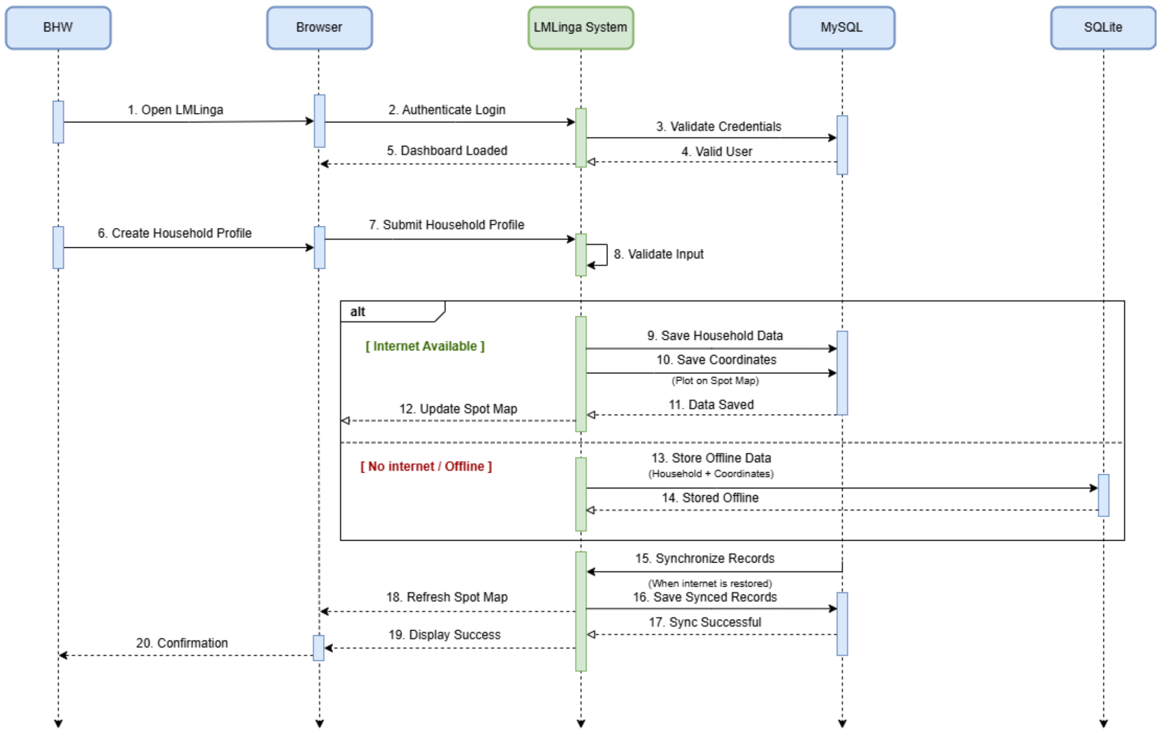

Figure 6. **Scenario**

Figure 6 shows the Scenario View of the proposed LMLinga system. The diagram illustrates the sequence of interactions among the Barangay Health Worker (BHW), Browser, LMLinga System, MySQL database, and SQLite database during the household profiling, automated spot mapping, and data storage processes. It demonstrates how the system manages user authentication, household data entry, validation, online and offline data storage, synchronization, and user confirmation.

The process begins when the Barangay Health Worker (BHW) accesses the LMLinga system through a web browser. The browser sends a login request to the LMLinga System, which verifies the user's credentials using the MySQL database. Once the credentials are successfully validated, the system grants access and displays the dashboard interface to the user. After logging in, the BHW creates a household profile by entering the necessary demographic and household information. The browser submits the profile data to the LMLinga System, where the information undergoes validation to ensure that all required fields are complete and accurate before being processed.

The diagram then presents an alternative flow based on internet connectivity. When an internet connection is available, the LMLinga System stores the household data in the MySQL database. The system also saves the geographic coordinates associated with the household, enabling automatic plotting on the digital spot map. After the data has been successfully stored, the system updates the spot map and reflects the newly added household location.

Alternatively, when internet connectivity is unavailable, the LMLinga System stores the household information and coordinates in the SQLite offline database. This offline functionality allows Barangay Health Workers to continue collecting data without interruption even in areas with unstable or limited internet access. The stored records remain locally available until connectivity is restored.

Once an internet connection becomes available, the SQLite database initiates the synchronization process by transferring the locally stored records to the LMLinga System. The system then saves the synchronized records to the MySQL database, ensuring that the central database remains updated with the latest household information. After successful synchronization, the system confirms that all records have been properly stored.

Finally, the LMLinga System sends a success notification to the browser, indicating that the profiling and synchronization processes have been completed successfully. The browser then displays a confirmation message to the Barangay Health Worker. Through this workflow, the diagram demonstrates the capability of LMLinga to support efficient household profiling, automated spot mapping, offline data collection, and reliable data synchronization, thereby improving the management and accessibility of community health information.

Figure 7. **Use Case Diagram**

Figure 7 shows the Use Case Diagram of the proposed LMLinga System, illustrating the interactions between the system and its primary users, namely the Resident, Barangay Health Worker (BHW), and Administrator. The diagram presents the major functionalities available within the system and identifies how each user interacts with specific features based on their roles and responsibilities.

For residents, the system provides functionalities that allow them to access services and information through the platform. These include registering an account, logging into the system, accessing the multilingual chatbot, submitting household representative requests, verifying one-time passwords (OTP), viewing household records, and viewing notifications. These features enable residents to communicate with the barangay health office, access household-related information, and receive important updates and announcements.

For Barangay Health Workers (BHWs), the system supports essential functions related to household and health data management. BHWs can register and log into the system, perform household profiling, conduct spot mapping activities, manage health records, generate reports, approve household representative requests, and view records. Through these functionalities, BHWs are able to collect and update household information, record health-related data, monitor community health conditions, and generate reports needed for decision-making and service delivery.

The Administrator has access to all functionalities available to Barangay Health Workers. This means that administrators can also perform household profiling, conduct spot mapping, manage health records, generate reports, approve household representative requests, and view records when necessary. However, the Administrator is provided with additional system management privileges that are not available to BHWs. Specifically, administrators can manage user accounts and deactivate BHW accounts to ensure proper access control, maintain system security, and oversee user activities within the platform.

The diagram also illustrates several supporting processes that are necessary for the execution of key system functions. Household profiling involves the storage of household information, spot mapping requires the recording of location data, health record management involves the storage of health-related information, and report generation requires the retrieval of database records. These supporting functions enable the successful completion of the system's primary operations and contribute to the overall efficiency of data management within the platform. Overall, the Use Case Diagram provides a high-level representation of the functional requirements of LMLinga. It demonstrates how residents, Barangay Health Workers, and administrators interact with the system to facilitate household profiling, automated spot mapping, health record management, multilingual communication, report generation, and user administration. Furthermore, it highlights the Administrator's broader access privileges, which encompass all BHW functionalities in addition to user management capabilities, thereby supporting effective system governance and operational oversight.

3.4 Software Development Stack

This section outlines the software technologies and platforms used in the development of the proposed system. It emphasizes the key components of the development environment that contribute to the system's functionality, performance, and maintainability. Table 5 presents the specifications of the software development stack utilized in building the proposed system.

Table 5: Software Development Stack

COMPONENT	TECHNOLOGY	VERSION
Operating System	Windows 11	25H2
Web Server	Apache	2.4.x
Back-end Language	PHP	11.0
Database Server	MySql	8.0
Front-end Language	HTML5, CSS3 (Bootstrap/Tailwind css), JavaScript	ES6+
Frameworks	Laravel	11.0
Chatbot	Ollama (Mistral Model)	v0.3
Spot Mapping	Leaflet.js	1.9
Deployment	Vercel	2024
Version Control	Github	2.44
Design	Figma	2024
APIs/Third-party Connection	iprogsms	1.0

As shown in Table 4, the proposed system utilizes a combination of modern and reliable software technologies to support its development and deployment. These components were carefully chosen to ensure that the system performs efficiently, remains scalable, and is easy to maintain over time. By integrating both front-end and back-end technologies, the system can provide a smooth and responsive user experience while maintaining strong data capabilities

The system operates on a Windows 11 operating system and uses Apache as its web server to handle client requests and deliver web content. On the back-end, PHP is used alongside the Laravel framework to handle the system's core functions, including data processing, validation, and workflow management. For the front-end, HTML5, CSS3, and JavaScript are implemented to create an interactive and user-friendly interface. Additionally, Leaflet.js is used to support map-based features, particularly for visualizing household data within the system.

For data management, MySQL serves as the database, ensuring that all records are organized, secure, and easily accessible. The system also includes a chatbot powered by the Ollama platform using the Mistral model, which enhances user interaction by providing automated assistance. In terms of development and deployment, the system is hosted using Vercel, with GitHub used for version control to efficiently manage code updates. Figma is used for designing the user interface, ensuring a consistent and well-structured layout. Furthermore, the integration of third-party APIs, such as IPROG SMS, allows for additional features like messaging services. Overall, these technologies work together to create a reliable, secure, and efficient system.

3.5 Hardware Development Stack

This section describes the hardware environment used to support the development, testing, and execution of the system. It highlights the essential computing components required to ensure that the system performs efficiently during coding, debugging, and deployment. Table 6 presents the hardware development stack specification, which reflects the capability of the device utilized throughout the development process.

Table 6: Hardware Development Stack Specification

COMPONENT	TECHNOLOGY	VERSION
Processor	AMD Ryzen 7 7445HS w/Radeon 740M Graphics	11^{th} Gen, 2.50 GHz

COMPONENT	TECHNOLOGY	VERSION
Installed RAM	16.0 GB	15.8 GB Usable
System Type	X64-based processor	

As shown in Table 6, the system is powered by an 11th Generation AMD Ryzen 7 7445HS w/Radeon 740M Graphics processor with a base speed of 2.50 GHz, which provides sufficient processing power to handle development tools, local servers, and multitasking activities. The device is equipped with 16 GB of RAM (15.8 GB usable), allowing smooth execution of multiple applications simultaneously, such as code editors, database management systems, and testing environments. The x64-based architecture enhances performance and supports modern applications that require higher memory capacity. These hardware specifications provide a stable and reliable environment for system development, enabling efficient processing, smooth multitasking, and effective testing throughout the project lifecycle.

3.6 Tools and Platforms

This section identifies the specific software applications, online platforms, and AI techniques utilized to develop, manage, and deploy the LMLinga system. These technologies ensure the efficient integration of household data management, spot mapping, and automated health screenings. Table 7 presents the development package specification that will be used in developing the system.

Table 7: Development Package Specification

PACKAGE / LIBRARY	VERSION	PURPOSE
Figma	2024	Interface design tool used to create UI wireframes and high-fidelity prototypes.
Visual Studio Code (VS Code)	1.115.0	Code editor utilized as the primary Integrated Development Environment (IDE).
XAMPP	8.2	Web server stack containing Apache and MySQL used for local development.
Apache HTTP Server	2.4	Web server used to process PHP scripts and serve web content.
HTML5, CSS3, JavaScript	ES6+	Core web technologies used to build the system’s frontend.
SQLite	3.46.0	Lightweight database for offline data capture and synchronization.

PACKAGE / LIBRARY	VERSION	PURPOSE
Leaflet.js	1.9.4	JavaScript library utilized for the interactive spot-mapping feature.
PHP and Laravel Framework	11.0	Primary backend PHP framework used to manage server-side logic and RBAC.
Ollama and Mistral	V0.3	Framework used to serve the Mistral LLM locally for data privacy.
RAG	N/A	AI technique implemented to enhance chatbot accuracy via health manuals.
IPROG SMS API	1.0	SMS gateway service utilized for secure resident authentication.
MySQL	8.0.35	Relational database management system used for structured health data storage.
Git and GitHub	2.44	Version control tools and repository management for collaborative development.
Vercel	2024	Cloud Platform-as-a-Service (PaaS) utilized for final system deployment.

As shown in Table 7, the proposed system utilizes various development packages and libraries to enhance system functionality and user interface design. Visual Studio Code (VS Code) serves as the primary development environment due to its extensive extension support for PHP and front-end debugging. Version control is managed through Git, with GitHub serving as the central repository to track changes and maintain code integrity. For local development and testing, XAMPP is utilized to simulate a production environment, allowing developers to run Apache, MySQL, and PHP locally before actual deployment.

The system incorporates specific frameworks and libraries to ensure a robust user experience. Laravel provides the backend structure and security for processing health data, while frontend technologies like HTML5, CSS3, and JavaScript manage the responsive layout and dynamic interactions. For specialized mapping features, Leaflet.js is used to visualize the geographic distribution of households. Furthermore, SQLite ensures data persistence by allowing health workers to save records locally in the browser when an internet connection is unavailable.

To facilitate automated health screenings, the system utilizes using Ollama Mistral as LLM to handle complex queries locally, maintaining resident privacy. Retrieval-Augmented Generation

(RAG) is further applied to ensure the chatbot provides fact-based answers derived from specific health manuals. Finally, secure account activation is facilitated by the IPROG SMS API, which delivers One-Time Passwords (OTPs) to verify resident identities.

3.7 Testing Plan

This section presents the testing procedures that will be used to make sure the proposed system works properly and meets the needs of the users. It will include the testing activities that will be done during the development of the system and the evaluation that will be conducted after implementation to check its functionality and usability.

3.7.1 Development Testing Plan

The development testing plan will outline the procedures that will be carried out during the system development phase to make sure that the individual components and integrated modules work according to the specified requirements. Figure 8 illustrates the development testing plan that will be used during system development.

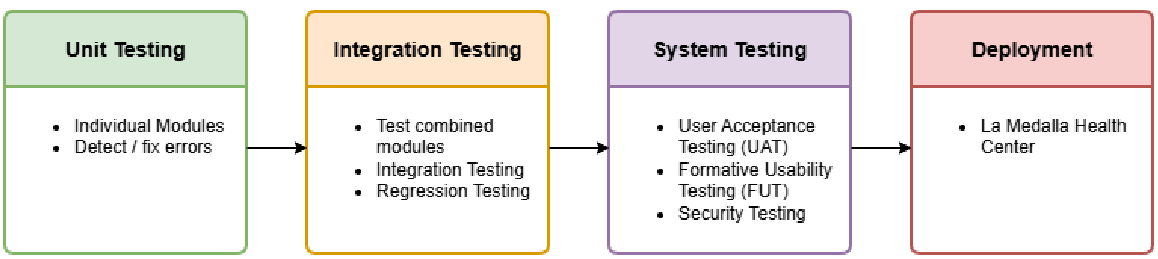

Figure 8. **Development Testing Plan Flowchart**

As shown in Figure 8, the development testing plan will have four stages to make sure the system is reliable before deployment. Unit testing will be done to check if each module works properly on its own. Integration testing will make sure that the modules work well together without data conflicts. System testing will evaluate the overall performance of LMLinga through usability, acceptance, and security testing. The system will only be deployed after all testing stages are completed.

User Acceptance Testing. User acceptance testing (UAT) for the LM Linga system will take place to prove the system meets the expectations and operational needs of both the Barangay Health Workers (BHW) and the residents of the barangay. The testing will ensure that the finished system meets all the user stories and acceptance criteria provided by the users during the requirements gathering part of the Extreme Programming (XP) development methodology. Barangay Health Workers who regularly perform household profiling, health record management, report generation, and spot-mapping activities will participate in the testing. The residents who will be using the system to register accounts, verify their OTP, use the chatbot, submit requests, and view information about their households will also participate in testing the functionality of the system relating to the residents. The participants will be performing predefined tasks and providing feedback regarding the accuracy, completeness, and appropriateness of the system's functions. The results of the UAT will determine if the modules developed meet the needs of the users and correspond to the actual workflow being performed at the Barangay health centre located in La Medalla. Any problems identified during the UAT process will be captured in writing and corrected before production release through multiple refinements. Table 8 shows the scenarios that the BHW will be performing during user acceptance testing.

Table 8: User Acceptance Testing

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
User Authentication Module		
Login Validation	Log in using valid credentials	User successfully accesses the appropriate dashboard
Household Profiling Module		
Add Household Record	Add a new household profile	Household profile is saved successfully
Add Resident Record	Add resident information to a household profile	Resident information is stored correctly
Add Health Record	Create a health record	Health record is saved and retrievable
Automated Spot Mapping Module		
Plot Household	Register household coordinates	Household is displayed on the digital map

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Report Generation Module		
Generate Report	Generate a household report	Report is generated accurately
Multilingual Chatbot Module		
Bikol-Iriga Query	Submit a health-related query in Bikol-Iriga	Chatbot provides a relevant response
Tagalog Query	Submit a health-related query in Tagalog	Chatbot provides a relevant response
English Query	Submit a health-related query in English	Chatbot provides a relevant response
Others		
Offline Synchronization	Save data offline and reconnect	Offline data synchronizes successfully
User Management	Create a new user account	User account is created successfully

Table 8 shows that UAT is being carried out to validate that the LMLinga system is capable of meeting the operational requirements and expectations of its intended end users. The UAT is comprised of extensive test cases for each major component of the LMLinga system such as user authentication, household profiling, automated spot mapping, report generation/multilingual chatbot services, offline synchronization, and user management. The UAT runs each test case in order to verify that each function works as it is supposed to (the correct output must occur). By passing all of these test cases, the LMLinga system will have satisfied the intended end-user requirements and therefore be ready to be implemented and used at La Medalla Barangay Health Center.

Integration Testing. Verification of the proper functioning of the individual modules will occur during integration testing to ensure that the combined components of the LMLinga system operate correctly as a complete application. Each of the system's modules will be executed as interconnected parts to enable the functioning of all components collectively. The system consists of numerous interconnected components, including digital household profiling, healthcare record management, automated report generation, GIS location mapping of homes, multilingual chatbots, and user authentication modules. Therefore, comparability must be made between all of the components with respect to communication and data transfer between said components. The researchers will conduct tests of interaction between the modules utilizing processes that involve executing scenarios involving multiple system functions. Some examples of these scenarios include verifying if the household profile created is reflected in the spot mapping module, confirming if the resident information is contained in the report being generated, and ensuring that the appropriate healthcare information is retrieved from the knowledge base for the user when using the chatbot. Integration testing will identify any errors in the use of the interface, discrepancies in data, communication failures, and problems operating between modules. Successful completion of the integration testing phase will provide evidence that the components of the system are functioning together cohesively, thereby allowing consistent workflows to operate uninterrupted through all component parts of the system. Table 9 shows the test scenarios that will be performed in integration testing.

Table 9: Integration Testing

INTEGRATED MODULES	DESCRIPTION	EXPECTED OUTPUT
Authentication + RBAC	Log in using different user roles	Appropriate permissions are applied
Household Profiling + Spot Mapping	Create household profile with coordinates	Household automatically appears on map
Household Profiling + Report Generation	Save profile then generate report	Newly added data appears in report
Household Profiling + Health Records	Add health record to a profiled resident	Health record is linked correctly
Chatbot + Knowledge Base	Submit a health inquiry	System retrieves relevant information
Offline Synchronization + Database	Synchronize offline records	Data uploads without duplication

As shown in Table 9, integration testing will be conducted to verify that the interconnected modules of the LMLinga system function correctly when operating as a unified application. The testing will focus on validating the interactions of the major components of the system, such as authentication/role-based access control, profiling households, mapping household locations, generating reports, managing health records, providing chatbot services, synchronizing with the offline database version, and connecting to the database. The test cases will assess whether there is an accurate exchange of data between the various modules and whether it has been processed without errors, duplicates, or any loss of data. Completing these tests successfully will indicate that the components of the system work together to deliver on the operational needs of the Barangay Health Center in La Medalla.

Regression Testing. During the entire development lifecycle, regression testing will be conducted to validate that the new functionality added (as a result of adding/ modifying/ fixing code) has no impact on the functionality of already existing system components. The methodology for development of this project will be through the Extreme Programming Methodology; therefore, continuous improvement and refinement of code is expected in each of the development increments. Anytime there is a change made to a particular part of the system, all of the modules that were previously tested for that particular part of the system will be retested through the execution of the same test cases that were used previously. Specifically, the modules that will be retested during regression testing include household profiling, resident record management, report generation, GIS-based spot mapping, chatbot responses, login authentication, and user account management. The purpose of performing regression testing is to maintain system stability and reliability and enable continued enhancements. Through the process of regression testing, the chance of introducing new defects into already functioning modules will be minimized, and the ability of the system to meet the requirements of the user and technical community will be maintained throughout the entire development process. Table 10 shows the test scenarios that will be performed during regression testing.

Table 10: Regression Testing

MODULE RE-TESTED	DESCRIPTION	EXPECTED OUTPUT
Authentication	Test login after updates	Login functions correctly
Household Profiling	Create profile after updates	Data is saved correctly
Health Records	Update health information	Changes are saved successfully
Spot Mapping	Plot household coordinates	Mapping functions correctly
Report Generation	Generate report after modifications	Reports generate accurately
Multilingual Chatbot	Submit chatbot inquiry	Chatbot responds correctly
Offline Synchronization	Synchronize offline records	Synchronization remains functional
User Management	Update user information	Updates are saved correctly
Security	Test role-based access after updates	Access permissions remain enforced

As shown in Table 10, regression testing will take place to confirm that the LMLinga system continues to work properly for previously tested functional modules after system updates, improvements, bug fixes, or changes made to the application. The tests will evaluate critical areas like authentication; household profile; management of health records; spot mapping; report generation; multi-language chatbot services; offline synchronization; user management; and security of user accounts. Each test case will ensure that the current functionality of each module remains stable and unchanged after the implementation of the most recent changes in the system. Completion of these tests will ensure that enhancements made to the product do not create additional defects and that the application's overall reliability and functionality are not compromised.

3.7.2 System Evaluation

Functionality Testing. The main objective of the functionality testing phase is to make sure that the LMLinga system works as one complete system, where every input gives the correct and expected output. In this phase, the system will be treated as a Black Box, which means the researchers will focus on how the system functions based on the given inputs and outputs instead of looking at the internal source code. Table 11 shows the test cases that the researcher will use to test the system performance.

Table 11: Functionality Testing Test Cases

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
	User Authentication Module	
Login Validation	Enter a valid username/password	User logs in successfully
Invalid Login Attempt	Enter incorrect credentials	Error message displayed
	Household Profiling Module	
Add Household Record	Input the head representative and details of the residents living in the household	Complete record saved
Edit Household Data	Modify existing household information	Updates saved correctly
	Automated Spot Mapping Module	
Generate Household Map	Create map from household addresses	Accurate geo-locations displayed
Map Export	Export spot map	PDF file generated
	Multilingual Chatbot Module	
Bikol-Iriga Query	Ask health information in Bikol-Iriga	Correct response in Bikol-Iriga
Filipino Query	Ask health information in Filipino	Correct response in Filipino
English Query	Ask health information in English	Correct response in English
	Report Generation Module	
Health Center Summary	Generate monthly report	Accurate report exported
Filter by categories	Filter reports by categories that the BHW wants	Filtered results displayed
	Input Validation Module	
Incomplete Form	Submit form with missing fields	Error prompts for required fields

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Invalid Data Entry	Enter invalid phone/email format	Validation error displayed
User Interface Module		
Role-Based Dashboard	Admin/BHW/Resident login views	Role-appropriate interface shown
Mobile Responsiveness	Access via mobile browser	Readable on 4G/3G devices
Household Representative Request Module		
Submit Request	The resident submits a request in the system	The request is saved successfully in the system
View Status	Resident checks the status of their submitted request	The correct and updated status is shown
Approval Process	The BHW approves or rejects the submitted request	The system displays the appropriate approval or rejection message
OTP Verification	The system sends an OTP after approval, and the resident uses it for verification	The OTP is received by the resident and works properly during verification
Security Module		
Data Encryption	Save Household data	AES-256 encryption applied
Unauthorized Access	Non-admin attempts admin functions	Access denied message

Based on Table 11, the functionality test cases will evaluate the performance of all LMLinga modules, including user authentication, household profiling, automated spot mapping, multilingual chatbot, report generation, input validation, user interface, and security. Each test case will check if the system responds properly to the user's actions by confirming whether the task is completed successfully or if the correct prompts and error messages are shown when invalid inputs are entered. The expected outputs will help determine if the system functions correctly and if all important features are working as intended for deployment at La Medalla Health Center.

Usability Testing. The main objective of the usability testing phase is to make sure that the LMLinga system is user-friendly, easy to navigate, and understandable for the users. In this phase, the system will be evaluated to identify problems, and the researchers will observe how users perform a specific task. The researcher will improve the user experience by basing decisions on actual user behavior to make sure that the system actually works for real users. Table 12 shows the test cases that the researcher will use to evaluate system usability.

Table 12: Usability Testing Test Cases

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
User Authentication Module		
Login Page Navigation	Access and navigate the login page	Login page is easy to locate and understand.
Invalid Login Attempt	User enters incorrect credentials	Clear and understandable error message is displayed
Household Profiling Module		
Add Household Record	User adds a household record using the form	Form is easy to understand and complete
Edit Household Record	User modifies existing household information	User can quickly locate and update fields
Automated Spot Mapping Module		
Generate Household Map	Generates a digital map from household addresses	Maps load quickly and easy to interpret
Map Export	User exports spot map	Export button is easy to find and use
Multilingual Chatbot Module		
Bikol-Iriga Query	User asks health information in Bikol-Iriga	Response is clear, natural and accurate in Bikol-Iriga term
Filipino Query	User asks health information in Filipino	Response is clear, natural and accurate in Filipino term
English Query	User asks for health information in English	Response is clear, natural and accurate in English term
Report Generation Module		
Health Center Summary	Generate reports	Steps are simple and intuitive
Category Filters	Filter reports by categories	Filters are easy to apply and understand
Input Validation Module		
Incomplete Form	User submit a form with missing fields	Helpful messages guide to complete the fields by the user
Invalid Data Entry Feedback	User enters incorrect data format on field	Error messages are clear and specific
User Interface Module		
Role-Based Dashboard	Different users (Admin/BHW/Resident) access their specific dashboard	Interface is organized and appropriate for user role

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Mobile Responsiveness	User access system via mobile device	Layout adapts well and remains easy to use
Household Representative Request Module		
Request Submission Ease	The resident fills out and submits the request form	The process is easy to follow and understand
Status Clarity	The resident views the status of their request	The status information is clear and easy to understand
Message Clarity	The resident receives approval or rejection messages	The message is clear and can be easily understood
OTP Experience	The resident receives and uses the OTP for verification	The OTP is received quickly and is easy to use
Security Module		
Data Security Transparency	User saves sensitive data	System reassures the user that data is secure
Unauthorized Access	Unauthorized user attempts restricted action	User is clearly informed that access is denied with a simple and understandable message

Based on Table 12, the usability test cases will evaluate the overall user experience of the LMLinga system, including user authentication, household profiling, automated spot mapping and multilingual chatbot, report generation, input validation, user interface, and security. Each test case will assess how easily users can understand, navigate, and interact with the system while performing a specific task. The expected output will help to determine if the system is user-friendly, efficient, and suitable for the users to effectively use at La Medalla Health Center.

3.7.3 Technical Evaluation

This section presents the technical validation test cases conducted by the IT Expert to evaluate the system architecture, security implementation, and performance capabilities of LMLinga. Table 13 shows the test cases that will be used in evaluating the technical aspect of the system.

Table 13: Technical Evaluation Test Cases

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Code Quality Review	Review the structure and organization of the Laravel framework used in the system.	The system should follow the MVC pattern and show proper separation of functions and components.
Database Design	Examine the MySQL database structure used in storing and managing system data.	The database should be properly normalized up to 3NF and should have appropriate indexing for better performance.
Scalability Assessment	Assess the system's ability to support multiple users at the same time.	The system should be able to handle 50 or more simultaneous BHW sessions without major issues.
OWASP Top 10 Compliance	Check the system for common web security vulnerabilities using ZAP scanning.	The system should not have any critical or high-level vulnerabilities.
Role-Based Access Control	Verify if user roles such as Admin, BHW, and Resident have proper access permissions.	Each user role should only be able to access the features assigned to them.
Data Encryption	Validate if sensitive household information is protected within the system.	Household data should be securely encrypted in the database.
JMeter Load Testing	Test the system performance under heavy user traffic using JMeter.	The system should maintain less than 2 seconds response time and 99% uptime.
Mobile Performance	Evaluate the system's performance when accessed using mobile devices and different network conditions.	The pages should load in less than 3 seconds on mobile browsers using 3G or 4G networks.
Leaflet.js Integration	Test the integration of the mapping feature used for household location plotting.	The system should display household locations accurately and in real time on the map.
RAG/Ollama Chatbot	Evaluate the chatbot's ability to generate multilingual responses.	The chatbot should respond in less than 1 second and support three different languages.
W3C Validation	Check if the system's HTML, CSS, and JavaScript follow web development standards.	The code should pass validation without errors.

TEST NAME	TEST DESCRIPTION	EXPECTED OUTPUT
Cross-Browser Testing	Test the system across different web browsers.	The system should function properly and consistently in Chrome, Firefox, Safari, and Edge.
API Documentation	Review the completeness of the system's API documentation.	The REST endpoints should be fully documented using Swagger or Postman.
User Manual Completeness	Check the completeness of the user guides prepared for different users of the system.	The manual should include clear step-by-step instructions for BHW, Admin, and Resident users.

Based on Table 13, these test cases will be conducted by the IT Expert to assess LMLinga in terms of system architecture, security implementation, performance capabilities, and standards compliance. Each test case was designed to determine whether the system meets the expected technical requirements and industry standards. These include the verification of the Laravel MVC structure, MySQL 3NF database design, OWASP security compliance, and the multilingual chatbot's performance under simulated peak conditions.

Security Testing. Security Testing is designed to assess whether the LMLinga system is capable of effectively protecting sensitive and confidential resident information from unauthorized access, breaches of data, and vulnerabilities inherent in common web applications. The system is designed to manage confidential household records and health records; therefore, data security and data privacy are among the paramount requirements.

The researchers plan to perform the security assessment using the Open Web Application Security Project (OWASP) Zed Attack Proxy (ZAP) as the main assessment tool for measuring Security. OWASP ZAP will be utilized to conduct Vulnerability Scans to determine if there are any potential vulnerabilities in the web application, such as an SQL Injection; Cross Site Scripting (XSS); Broken Authentication; Insecure Session Management; Security Misconfiguration, and other types of Common Web Application Vulnerabilities.

In addition to conducting vulnerability scans, the researchers will also perform testing on Role-Based Access Control (RBAC) mechanisms that have been implemented within the LMLinga system to ensure only those authorized users who have been assigned permission can access protected information. The results from the security assessment will be analyzed and acted upon to ensure compliance with applicable data privacy and software security requirements. Table 14 shows the scenarios that will be performed during security testing.

Table 14: Security Testing

MODULE	SECURITY TEST	EXPECTED RESULT
Authentication	Brute-force login attempt	Unauthorized access is prevented
Authentication	Session management test	Session expires properly after logout
RBAC	Access restricted pages	Access is denied to unauthorized users
Household Profiling	OWASP ZAP SQL Injection Scan	No SQL injection vulnerability detected
Authentication	OWASP ZAP Cross-Site Scripting (XSS) Scan	No XSS vulnerability detected
Multilingual Chatbot	Input validation test	Malicious input is sanitized
Report Generation	Unauthorized report access test	Access is denied
User Management	Privilege escalation test	Users cannot elevate privileges
Database	Sensitive data exposure test	Sensitive information remains protected
Entire System	OWASP ZAP Vulnerability Scan	No Critical or High vulnerabilities detected

As shown in Table 14, security testing will be completed to determine whether the LMLinga system can adequately secure household and health data from unauthorized access, malicious attacks, and security risks. Security Testing will also include functional tests to assess the following six key security areas of the system, such as authentication, session management, role-based access control, input validation, protection of data, and OWASP ZAP vulnerability assessments. Each test case has been designed to validate that selected security controls exist and are working correctly on the various modules in the solution. Completion of all security tests will show that the system can protect user data, enforce access restrictions on users, and assure the confidentiality, integrity, and security of data within the solution.

Performance Testing. Performance Testing will be done to test the responsiveness, stability, and reliability of the LMLinga system when it is delivering the expected load to the concurrent users accessing the various system modules (authentication, household profiling, spot mapping, report generation, and chatbot services). The goal of this testing is to ensure that as multiple users access the LMLinga system simultaneously through the above-stated modules, the system maintains an acceptable level of performance. Performance Testing will generate "real-world" types of usage scenarios by sending concurrent requests and transactions across all system components. The Performance Testing results will be used to identify existing bottlenecks; measure the system response times; and confirm that the system can support the expected operating conditions for the Barangay Health Center of La Medalla with no data loss, no delays in processing, and no interruptions of service. Table 15 shows the performance tests to be conducted during performance testing.

Table 15: Performance Testing

Module	Performance Test	Expected Result
Authentication	Simulate 50 concurrent logins	System responds within acceptable response time
Household Profiling	Simultaneous profile submissions	Records are saved without errors
Spot Mapping	Concurrent map loading requests	Map loads successfully
Report Generation	Multiple report generation requests	Reports are generated correctly
Multilingual Chatbot	Multiple chatbot queries	Chatbot responds without failure
Database	Simultaneous database transactions	No data loss or corruption occurs
Entire System	Load test with concurrent users	System remains stable under expected workload

As shown in Table 15, Performance Testing will be conducted to assess the responsiveness, stability, and reliability of the LMLinga system under concurrent user activities and expected operational workloads. Performance Testing includes all of the major system modules (authentication, household profiling, spot mapping, report generation, multilingual chatbot services, and database operations). Each performance test case has been designed to determine whether the system can successfully process multiple simultaneous requests while maintaining acceptable response times, data integrity, and functionality of the system. Successful completion of each performance test case will validate the system's ability to effectively support the day-to-day operations of the health center while operating at a level of performance consistent with anticipated volume of use.

3.8 System Failure Plan

The LMLinga web application is designed to allow for the completion of a variety of tasks by three types of users (Administrator, Barangay Health Worker (BHW), and Resident) and was created using the Vercel platform. Laravel/ PHP serves as the backend language, while MySQL is used to store all data in the primary database, and SQLite will be used locally by BHWs to perform offline work. Leaflet.js is used for GIS spot mapping, while Ollama (Mistral) will provide local language model (LLM) chatbot inference. For health-related data retrieval, RAG will be used, and the IPROG SMS API will provide OTP delivery to resident users for household representative requests. Table 16 shows the system failure plan for LMLinga.

Table 16: System Failure Plan

DESCRIPTION/ FUNCTION	SYSTEM FAILURE SCENARIO & PLAN
USER AUTHENTICATION MODULE	
This module is where all users, like Admin, BHW, and Resident, will log in to the system. Its main function is to check if the username and password are correct. If the	**Scenario: Credential Mismatch** This occurs when a user enters an incorrect username or password. If authentication fails, the user cannot bypass the login screen to reach their role-specific dashboard (Administrator, BHW, or Resident). For Administrator and BHW, this prevents to access accessing records of the

<table>
<tr><td>user enters the wrong credentials, the system will show an error message to avoid unauthorized access.</td><td>residents. For residents, this prevents to access chatbots and their health records.
Preventive Measures:
1. The system checks all login fields for the correct format before processing the authentication request.
2. Ensures that once the gate is passed, users are only directed to the interface appropriate for their role.
3. The system is programmed to display clear, understandable error messages to the user if credentials do not match, preventing unauthorized entry while informing the user of the mistake.
Recovery Procedures:
1. If wrong credentials are entered, the system immediately blocks entry and displays an "Invalid Login Attempt" error message.
2. The user is prompted to re-enter their credentials.
3. For BHW, the Administrator uses the Manage Users module to verify account status, reset passwords, or re-activate accounts as necessary.
Backup Method:
The system relies on the Administrator's oversight of the CentralDB to resolve manual login issues.</td></tr>
<tr><td colspan="2">HOUSEHOLD PROFILING MODULE</td></tr>
<tr><td>This module is used for recording and managing household health information in digital form. Users (BHW) can add new household records, including the head of the family and other members. It also allows editing or updating of existing data using forms that are simple and easy to understand.</td><td>Scenario: Loss of Connectivity during Field Profiling
This occurs when a BHW is in an area of the barangay with unstable or limited internet access while trying to save or update household demographics. Without a fallback, BHWs would be unable to save records to the database, potentially leading to incomplete community health data or the need to revert to manual, paper-based profiling.
Preventive Measures:
1. To ensure offline data capture, the system integrates SQLite as a lightweight browser-based database for storing data when an internet connection is unavailable.
2. The system is programmed to check for internet availability during every data processing request
3. The Input Validation Module ensures all fields are accurate before they are saved to either the local or cloud database to prevent corrupted records.
Recovery Procedures:
1. The LMLinga system detects the lack of internet connection and automatically redirects the profiling data to the local SQLite database.</td></tr>
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<tr><td></td><td>
2. The BHW continues to enter or edit household records without interruption. The system saves these records and their geographic coordinates locally.
3. Once an internet connection becomes available, the system initiates a background synchronization process.
4. The system transfers the records from SQLite to the MySQL central database and provides a success notification once the records are updated.
Backup Method:
The system maintains SQLite on the user's device as an immediate local backup for all field-collected data, ensuring that the CentralDB (MySQL) eventually reflects all records regardless of connectivity status during the initial entry.
</td></tr>
<tr><td colspan="2">AUTOMATED SPOT MAPPING MODULE</td></tr>
<tr><td>This module is responsible for showing the location of households in a digital map. It uses geographic data to automatically plot the households. It also allows the user to export the map into a PDF file for reporting purposes.</td><td>
Scenario: Map Visualization Failure Tile Loading Error
It occurs when the Leaflet.js library fails to initialize, or used to show the actual map background cannot be fetched due to network restrictions or platform downtime. Barangay Health Workers (BHWs) and Administrators cannot view the real-time plotting of households on the interactive map. Furthermore, the inability to render the map prevents the Map Export feature from creating the PDF files required for health center reporting.
Preventive Measures:
1. The system is designed to store geographic coordinates (Latitude and Longitude) as raw data in the MySQL or SQLite database, ensuring the data exists even if the map cannot be displayed.
2. The system utilizes SQLite for offline data handling, allowing coordinates to be captured and saved locally when internet connectivity for live map tiles is unavailable.
3. Continuous testing of Leaflet.js integration to ensure that household locations plot accurately and in real-time under standard operating conditions.
Recovery Procedures:
1. Access the database to confirm that the geographic coordinates for the households are present and uncorrupted.
2. Refresh the browser or clear the application cache to force the Leaflet.js library to re-load and attempt to re-establish a connection with the map tile provider.
3. Navigate to the BHW or Administrator dashboard and manually trigger the Generate Household Map
</td></tr>
</table>

	command to re-plot coordinates once the visualization layer is confirmed active. 4. Execute the Map Export function immediately after the map renders to produce a PDF file, serving as a physical backup for the health center's records. **Backup Plan:** The primary backup for this module is the relational coordinate records stored within the MySQL and SQLite databases. Because the system records location data as a supporting process for household profiling, the geographic "source of truth" remains accessible for manual reporting even if the automated digital map fails.
MULTILINGUAL CHATBOT MODULE	
This module acts like a 24/7 assistant that gives health information to the users. It can respond in English, Filipino, and Bikol-Iriga. It uses RAG so that the answers are based on reliable sources like DOH documents.	**Scenario: AI Service Failure, Local Resource Exhaustion, or Knowledge Index Failure** This occurs if the local hosting platform (Ollama) stops responding due to hardware limitations (RAM/CPU) or if the RAG component fails to retrieve data from the health manual datasets, leading to a "hallucination" or a total failure to generate a response. Residents are unable to get immediate answers to health-related queries in their preferred language. **Preventive Measures:** 1. Implement a screening layer to identify language parameters and ensure the query is relevant (e.g., matching keywords like "fever" or "cough") before the request even reaches the Large Language Model, reducing unnecessary system load. 2. Implement strict behavioral restrictions that prevent the model from straying into unrelated topics or providing ungrounded information. **Recovery Procedures:** 1. Manually restart the local Ollama hosting environment to clear system memory and re-initialize the Mistral LLM 2. Verify the connection to the indexed health manuals and DOH documentation to ensure the AI can once again pull factual information. 3. Conduct test queries in English, Filipino, and Bikol-Iriga to confirm the rule-based filter is correctly identifying the language and triggering the RAG process. **Backup Method:** The primary backup for the AI's generative responses is the Static Dataset of Health Manuals. If the conversational AI

<table>
<tr><td></td><td>remains unstable, the system can provide direct links or access to the digital PDF versions of the DOH health documents used to train the RAG model, ensuring residents still have access to reliable health information.</td></tr>
<tr><td colspan="2">REPORT GENERATION MODULE</td></tr>
<tr><td>This module is used to automatically generate reports from the collected household data. For example, it can create monthly reports. It also has filter options so the user can view specific data depending on what they need.</td><td>Scenario: Record Retrieval Failure
This occurs when the system is unable to successfully fetch the required records from the database due to high server load or complex filtering requests during report generation. Barangay Health Workers (BHWs) and Administrators are unable to view filtered results or export the Health Center Summary into a PDF format. This halts the reporting workflow necessary for monitoring community health conditions.
Preventive Measures:
1. The MySQL database is designed with 3NF normalization and proper indexing to enhance retrieval performance and prevent timeouts
2. Ensuring that only valid filter parameters are processed to avoid resource-heavy "broken" queries.
Recovery Procedures:
1. Check the status of the MySQL database within the Data Layer to ensure it is responsive and that no table locks are preventing data retrieval.
2. Navigate to the Report Generation Module and clear all active filters to re-attempt a basic "All Records" summary to identify if a specific category is causing the failure.
3. Once the connection is stable, re-apply the desired category filters through the BHW or Admin dashboard to fetch the specific data needed.
4. After the results are displayed on the interface, immediately execute the Export Report function to generate a PDF file, providing a permanent digital and physical backup of the data.
Backup Method:
If the automated reporting interface fails, raw data can be retrieved directly from the MySQL central database or the SQLite local database (if synchronized) to manually compile the health center summaries, ensuring that community data is never lost.</td></tr>
<tr><td colspan="2">USER INTERFACE MODULE</td></tr>
<tr><td>This module focuses on how the system looks and how the users interact with it. It</td><td>Scenario: Front-End Application Failure or Vercel Service Disruption</td></tr>
</table>

provides different dashboards depending on the role of the user (Admin, BHW, Resident). It is also mobile responsive so it can still be used in phones and tablets.	This occurs if the CSS/JavaScript files fail to load properly, causing the layout to break, or if the Vercel cloud platform experiences an outage, making the web-based dashboards inaccessible to all users. Users are unable to interact with the system's "gate" (the login page) or access their specific dashboards. If the mobile responsiveness fails, BHWs in the field using tablets or phones will experience unreadable interfaces, halting mobile profiling and mapping activities. **Preventive Measures:** 1. Use of GitHub for repository management, allowing for immediate identification of code changes that may have broken the UI. 2. Regular validation across Chrome, Firefox, Safari, and Edge to ensure consistent UI behavior. 3. Technical requirements ensure that pages load in less than 3 seconds on mobile browsers (3G/4G) and follow W3C standards to prevent cross-browser rendering issues. **Recovery Procedures:** 1. Confirm if the failure is platform-wide by checking the Vercel status page or if it is a build-specific error in the Application Layer. 2. Use Git/GitHub to roll back the deployment to the most recent stable version of the interface that passed all usability and functionality tests. 3. After restoration, verify that the Mobile Responsiveness is intact by testing the interface on multiple mobile devices to ensure readability on 3G/4G networks. 4. Ensure that the Role-Based Access Control (RBAC) logic correctly directs Admin, BHW, and Resident users to their respective dashboards upon passing the authentication gate. **Backup Method:** The backup for the UI is the GitHub version history, which allows for near-instant restoration of the presentation layer.
HOUSEHOLD REPRESENTATIVE REQUEST MODULE	
This module allows residents to send a request to access their household information. It handles the submission of requests, tracking of status, and approval by the BHW. It	**Scenario: OTP Delivery Failure (SMS Signal Blackout)** This occurs when a resident's request to access their household information that will be approve by a Barangay Health Worker (BHW), but the IPROG SMS API cannot deliver the required verification code because the resident is in a "no signal" area or the SMS gateway is experiencing downtime. Identity verification is the first step, a failure here prevents the resident

also uses OTP for verifying the identity of the resident.	from successfully submitting their request. The Barangay Health Worker (BHW) will never see the request in their dashboard for approval, resulting in a critical failure of the resident's ability to access household health records. **Preventive Measures:** 1. To ensure reliable OTP delivery, the system is configured in the Laravel application layer to use IPROG (SMS) as the primary channel and Gmail (Email) as a secondary channel when SMS services are unavailable. 2. Mandatory registration of a secondary contact method (Gmail address) for every resident to act as a failover channel. **Recovery Procedures:** 1. The system monitors the IPROG SMS API. If it detects a failure or the resident clicks "Resend Code" due to signal loss, the system immediately offers the "Receive Code via Gmail" option. 2. Upon the resident's selection, the system dispatches the verification code via Gmail SMTP to the registered email address. 3. The resident enters the email-received code, which is verified by the Input Validation Module against the CentralDB to confirm their identity. 4. Once the OTP is successfully verified, the request is officially submitted and appears on the BHW's dashboard as a "Verified Request" ready for final administrative approval or rejection. **Backup Method:** The system relies on Gmail SMTP as a backup to the IPROG SMS API, ensuring that the first gate of the request process (Identity Verification) is always functional.
SECURITY MODULE	
This module makes sure that all sensitive data is protected. It uses AES-256 encryption to secure the household data and applies role-based access so only authorized users can access some features. OTP is also used as an additional security measure for verification.	**Scenario: Unauthorized Privilege Escalation or Encryption Key Compromise** This occurs if a non-admin user (Resident or external actor) attempts to bypass RBAC to access administrative functions, or if the system fails to correctly decrypt household records due to a key integrity error. If RBAC fails, unauthorized users could modify health records or deactivate BHW accounts. If encryption fails, sensitive health data becomes unreadable or, in a worst-case breach, exposed in plain text, violating the privacy of the residents and their households. **Preventive Measures:**

	1. Continuous use of OWASP ZAP to identify and patch high-level web application security risks and Top 10 vulnerabilities. 2. Ensuring all household data is stored using AES-256 encryption in the CentralDB (MySQL) and the SQLite offline database. 3. Hardcoding Role-Based Access Control within the Application Layer (Laravel) to ensure that every request is validated against the user's assigned role (Admin, BHW, or Resident) before execution. **Recovery Procedures:** 1. The system must automatically detect and block any user session that triggers an "Unauthorized Access" error more than three times consecutively. 2. Use OWASP ZAP logs and database access logs to identify the point of entry and determine if any sensitive data was viewed or modified during the breach. 3. Verify the status of the AES-256 encryption keys; if a key is compromised, initiate an emergency key rotation to re-encrypt sensitive data in the MySQL and SQLite databases. 4. Perform a system-wide check of the RBAC permissions to ensure that Admin, BHW, and Resident roles are correctly restricted to their specific modules as defined in the Logical View. **Backup Method:** The system utilizes AES-256 encrypted database snapshots as its primary backup and recovery method to ensure data integrity, while the Laravel framework enforces a secure MVC structure with Role-Based Access Control (RBAC) to maintain proper separation of concerns and serve as a structural safeguard against security logic bypass and unauthorized access.

As shown in Table 16, the System Failure Plan outlines the potential failure scenarios that may occur within the major modules of the LMLinga system, together with the corresponding preventive measures, recovery procedures, and backup methods. It includes preventative measures to stop failures from occurring, as well as recovery processes to restore failed modules, and backup options to recover lost data. The System Failure Plan is also a contingency plan for maintaining the LMLinga system in an operational state while minimizing impacts due to technical issues, connectivity issues, service interruptions, security vulnerabilities, or application failures. Each of the major modules of the LMLinga system were rated on their criticality, then analyzed to determine potential points of failure and possible responses necessary for maintaining the integrity of data, availability of systems, and reliability of services. By implementing these strategies to manage possible system failures, the LMLinga system can meet the operational requirements of La Medalla Barangay Health Center while continuing to provide critical health information services.

3.9 System Deployment

This section describes how the proposed system will be deployed within the Barangay Health Center of La Medalla. It talks about where and how the researchers will deploy the system, and what has to happen in order to successfully implement it for actual users.

3.9.1 Deployment Environment

The deployment of LMLinga will be a web-based application hosted on Vercel. Users will access the application on various devices with internet connectivity (e.g., smartphones, tablets, laptops) using web browsers. The application will use MySQL as its primary database for creating and managing household profiles, health records, spot mapping, generating reports, and storing user accounts. To facilitate offline data storage during the house-to-house profiling process, SQLite will be integrated into LMLinga so that Barangay Health Workers can continue collecting data when the internet is not accessible and later synchronize the data to the relevant records when an internet connection is available. The following external services will also be implemented in the deployment environment: IPROG SMS API to allow for user verification using one-time passwords (OTP), and Ollama to create a multilingual chatbot. To ensure data confidentiality, integrity, and security, the deployment environment will utilize AES-256 encryption and implement role-based access control over system data.

3.9.2 Deployment Procedure

After the successful development and testing phases of the application that has been created in Laravel, the actual production application will be deployed into the designated host environment and connected to its databases and other external services. The process of deploying the application will first require the configuration of the application's database as well as creating users and setting roles and permissions, after which existing household data and any attached data will either be entered in or imported into the system's database to establish the initial data set for the application to operate from. Following the pilot deployment phase, the researchers will initiate pilot deployment activities within the Barangay Health Center to ensure that all modules, integrations, and data synchronization functions as expected within the actual operating environment. At this stage, the researchers will observe the performance of the application, determine technical issues with the application, and make appropriate corrections to the application before the full implementation of the application.

3.9.3 User Orientation and Full Implementation

Before the complete implementation, the system will be introduced to the program administrators and the Barangay Health Workers through orientation and training sessions. The purpose of these training activities is to familiarize users with the features of the system, its workflows, and its operational procedures. The specific training topics to be addressed include user authentication, household profiling, spot mapping, management of health records, report generation, and using the multilingual chatbot. During the pilot of LMLinga, input was received from users through their experiences during the pilot deployment and training sessions. This input has been reviewed to identify potential areas for improvement to ensure that the system meets user requirements. Once all identified issues have been resolved and the system has passed all specified functionality, usability, security, and performance testing, LMLinga will then be fully deployed and used as the central repository for household profiling, spot mapping, health record management, and health information services for the Barangay Health Center of La Medalla.

Notes

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